trabalho1_experimentos

February 22, 2026

1 Experimentos do Trabalho 1 - Reconhecimento de Padrões

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1.0.1 1. Imports e Carregando Dataset

[2]:
```
import time
import warnings

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from scipy import stats
from sklearn.decomposition import PCA
from sklearn.metrics import accuracy_score, classification_report,␣
↪confusion_matrix
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from ucimlrepo import fetch_ucirepo

from classificao_padroes.models_trabalho1 import KNN, MAXCO, MDC

warnings.filterwarnings('ignore')
```

```
2025-12-14 19:42:05.638 | INFO     |
classificao_padroes.config:<module>:11 -
PROJ_ROOT path is: /home/apo-pc/Documents/Github/Trabalhos_Rec_Padroes
```

[3]:
```
# Carregar o dataset
vertebral_column = fetch_ucirepo(id=212)
X = vertebral_column.data.features.to_numpy()
y_raw = vertebral_column.data.targets.to_numpy().ravel()

# Codificar as labels para valores numéricos
le = LabelEncoder()
y = le.fit_transform(y_raw)
```

```
print("Shape of X:", X.shape)
print("Shape of y:", y.shape)
print("Classes:", le.classes_)
print("Encoded Classes:", np.unique(y))
```

```
Shape of X: (310, 6)
Shape of y: (310,)
Classes: ['Hernia' 'Normal' 'Spondylolisthesis']
Encoded Classes: [0 1 2]
```

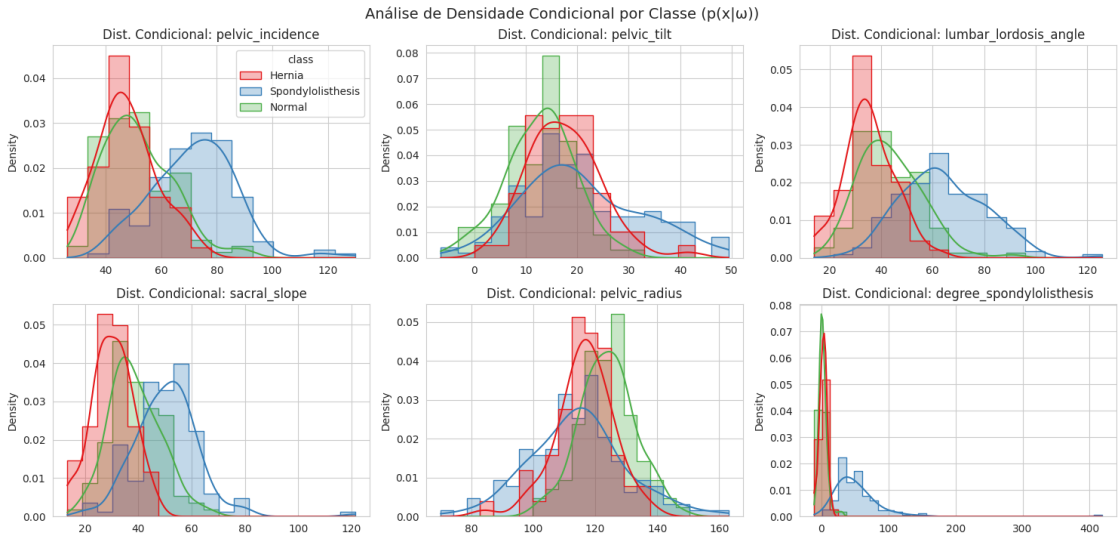
1.2 Análise dos Dados

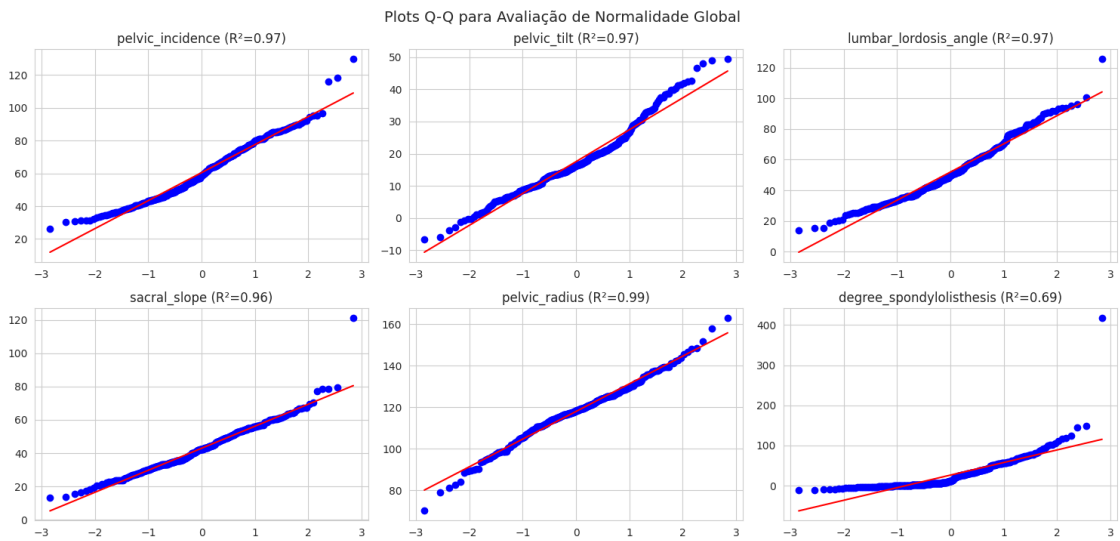
[4]:
```
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import math

# Configurações estéticas para publicação
sns.set_style("whitegrid")
plt.rcParams.update({'font.size': 10, 'figure.titlesize': 14, 'axes.titlesize':␣
↪12})

def plot_compact_analysis(df, features, target_col='class'):
    """
    Gera visualizações compactas adequadas para relatórios técnicos.
    Separa a análise em duas figuras: Distribuições e Normalidade (Q-Q).
    """
    n_features = len(features)
    n_cols = 3
    n_rows = math.ceil(n_features / n_cols)

    # --- FIGURA 1: Distribuições Condicionais (Histogramas/KDE) ---
    fig_hist, axes_hist = plt.subplots(n_rows, n_cols, figsize=(15, 3.5 *␣
↪n_rows))
    axes_hist = axes_hist.flatten() # Facilita a iteração

    for i, col in enumerate(features):
        ax = axes_hist[i]

        # Plotando as densidades condicionais (p(x|w_i))
        sns.histplot(
            data=df, x=col, hue=target_col,
            kde=True, element="step", stat='density',
            common_norm=False, ax=ax, palette='Set1', alpha=0.3
        )
        ax.set_title(f'Dist. Condicional: {col}')
        ax.set_xlabel('')
```

```python
        # Remove a legenda de todos exceto o primeiro para economizar espaço
        if i > 0 and ax.get_legend() is not None:
            ax.get_legend().remove()

    # Remove eixos vazios se houver
    for j in range(i + 1, len(axes_hist)):
        fig_hist.delaxes(axes_hist[j])

    plt.tight_layout()
    plt.suptitle("Análise de Densidade Condicional por Classe (p(x| ))", y=1.02)
    plt.show()

    # --- FIGURA 2: Plots Q-Q (Avaliação de Normalidade Global) ---
    fig_qq, axes_qq = plt.subplots(n_rows, n_cols, figsize=(15, 3.5 * n_rows))
    axes_qq = axes_qq.flatten()

    shapiro_results = []

    for i, col in enumerate(features):
        ax = axes_qq[i]

        # Q-Q Plot
        (osm, osr), (slope, intercept, r) = stats.probplot(df[col],␣
↪dist="norm", plot=ax)
        ax.set_title(f'{col} (R²={r**2:.2f})')
        ax.set_xlabel('')
        ax.set_ylabel('')

        # Teste estatístico (apenas para registro na tabela depois)
        stat, p_value = stats.shapiro(df[col])
        shapiro_results.append({
            'Atributo': col,
            'W-Stat': f"{stat:.3f}",
            'p-value': f"{p_value:.3e}",
            'H0 (Normal)': "Não Rejeita" if p_value > 0.05 else "Rejeita"
        })

    # Remove eixos vazios
    for j in range(i + 1, len(axes_qq)):
        fig_qq.delaxes(axes_qq[j])

    plt.tight_layout()
    plt.suptitle("Plots Q-Q para Avaliação de Normalidade Global", y=1.02)
    plt.show()

    return pd.DataFrame(shapiro_results)
```

```python
# --- Execução ---
 # 2. Estatísticas Descritivas

 # 1. Carregamento e Preparação dos Dados

print("--- Carregando Dataset ---")

vertebral_column = fetch_ucirepo(id=212)

df = vertebral_column.data.features.copy()
features = df.columns.tolist()

# Adicionando o target ao dataframe para facilitar visualizações

df['class'] = vertebral_column.data.targets.to_numpy().ravel()

display(df.describe().T)

print("\n=== 3. Análise Visual Compacta para Relatório ===")
df_shapiro = plot_compact_analysis(df, features, target_col='class')

print("\n=== Teste de Normalidade (Shapiro-Wilk) ===")
# Dica: Em relatórios, prefira exibir p-values muito pequenos como < 0.001
display(df_shapiro)
```

```
--- Carregando Dataset ---
                          count        mean        std        min         25%  \
pelvic_incidence          310.0   60.496653  17.236520  26.147921   46.430294
pelvic_tilt               310.0   17.542822  10.008330  -6.554948   10.667069
lumbar_lordosis_angle     310.0   51.930930  18.554064  14.000000   37.000000
sacral_slope              310.0   42.953831  13.423102  13.366931   33.347122
pelvic_radius             310.0  117.920655  13.317377  70.082575  110.709196
degree_spondylolisthesis  310.0   26.296694  37.559027 -11.058179    1.603727

                                 50%         75%         max
pelvic_incidence           58.691038   72.877696  129.834041
pelvic_tilt                16.357689   22.120395   49.431864
lumbar_lordosis_angle      49.562398   63.000000  125.742385
sacral_slope               42.404912   52.695888  121.429566
pelvic_radius             118.268178  125.467674  163.071041
degree_spondylolisthesis   11.767934   41.287352  418.543082

=== 3. Análise Visual Compacta para Relatório ===
```






```
=== Teste de Normalidade (Shapiro-Wilk) ===
```

	Atributo	W-Stat	p-value	H0 (Normal)
0	pelvic_incidence	0.971	7.132e-06	Rejeita
1	pelvic_tilt	0.966	1.321e-06	Rejeita
2	lumbar_lordosis_angle	0.972	9.221e-06	Rejeita
3	sacral_slope	0.964	5.887e-07	Rejeita
4	pelvic_radius	0.989	1.661e-02	Rejeita
5	degree_spondylolisthesis	0.697	2.908e-23	Rejeita

1.0.2 2. Definição e Execução dos Experimentos

[5]:
```
def run_experiments(X, y, test_size, n_runs=100):
    le = LabelEncoder()
    y = le.fit_transform(y.ravel())

    minkowski_orders = [0.5, 2/3, 1, 3/2, 2, 5/2]
    classifiers = {f'KNN (m={m:.2f})': KNN for m in minkowski_orders}
    classifiers['MDC'] = MDC
    classifiers['MDC Robusto'] = (MDC, {'robustVersion': True})
    classifiers['MAXCO'] = MAXCO

    results = {name: [] for name in classifiers.keys()}

    for i in range(n_runs):
        X_train, X_test, y_train, y_test = train_test_split(X, y,␣
 ↪test_size=test_size, random_state=i, stratify=y)

        for name, clf_class in classifiers.items():
            params = {}
            if isinstance(clf_class, tuple):
                clf_class, params = clf_class

            # Treinamento
            start_train_time = time.time()
            if 'KNN' in name:
                m = float(name.split('=')[1][:-1])
                model = clf_class(X=X_train, Y=y_train, k=1, minkowskiOrder=m)
            elif 'MDC' in name:
                model = clf_class(X=X_train, Y=y_train, **params)
                model.train()
            elif 'MAXCO' in name:
                model = clf_class(X=X_train, Y=y_train)
                model.train()
            train_time = time.time() - start_train_time

            # Teste
            start_test_time = time.time()
            y_pred = model.predict(X_test)
            test_time = (time.time() - start_test_time) / len(X_test)

            # Salvar resultados
            accuracy = accuracy_score(y_test, y_pred)
            results[name].append({
                'run': i,
                'accuracy': accuracy,
                'train_time': train_time * 1000,
```

```
                'test_time': test_time * 1000,
                'y_test': y_test,
                'y_pred': y_pred
            })
    return results, le
```

1.0.3 3. Análise dos Resultados

[6]:
```
def analyze_results(results, le, train_ratio):
    summary_data = []
    for name, runs in results.items():
        accuracies = [r['accuracy'] for r in runs]
        train_times = [r['train_time'] for r in runs]
        test_times = [r['test_time'] for r in runs]

        # OBS 1 & 2
        summary = {
            'Classifier': name,
            'Train Ratio': train_ratio,
            'Acuracia Media': np.mean(accuracies),
            'Std Acuracia': np.std(accuracies),
            'Min Acuracia': np.min(accuracies),
            'Max Acuracia': np.max(accuracies),
            'Median Acuracia': np.median(accuracies),
            'Tempo de Treino Medio (ms)': np.mean(train_times),
            'Tempo de Treino Std (ms)': np.std(train_times),
            'Tempo de Teste Medio (ms)': np.mean(test_times),
            'Tempo de Teste Std (ms)': np.std(test_times),
        }
        summary_data.append(summary)

    summary_df = pd.DataFrame(summary_data).sort_values(by='Acuracia Media',␣
↪ascending=False)
    display(summary_df.drop(columns=['Train Ratio']))

    # OBS 2: Matrizes de confusão para melhor e pior rodada
    for name, runs in results.items():
        accuracies = [r['accuracy'] for r in runs]
        best_run_idx = np.argmax(accuracies)
        worst_run_idx = np.argmin(accuracies)
        best_run = runs[best_run_idx]
        worst_run = runs[worst_run_idx]

        print(f'--- {name} ---')
```

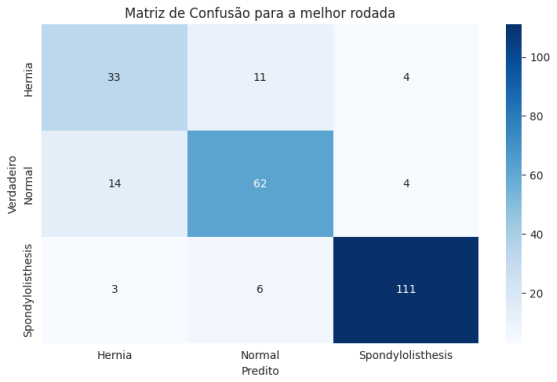
```
        print(f'Melhor rodada (#{best_run["run"]}) Acurácia:␣
↪{best_run["accuracy"]:.4f}')
        print(f"Pior rodada (#{worst_run['run']}) Acurácia:␣
↪{worst_run['accuracy']:.4f}")

        fig, axes = plt.subplots(1, 2, figsize=(15, 5))
        cm_best = confusion_matrix(best_run['y_test'], best_run['y_pred'])
        sns.heatmap(cm_best, annot=True, fmt='d', cmap='Blues', xticklabels=le.
↪classes_, yticklabels=le.classes_, ax=axes[0])
        axes[0].set_title(f'Matriz de Confusão para a melhor rodada')
        axes[0].set_xlabel('Predito')
        axes[0].set_ylabel('Verdadeiro')

        cm_worst = confusion_matrix(worst_run['y_test'], worst_run['y_pred'])
        sns.heatmap(cm_worst, annot=True, fmt='d', cmap='Reds', xticklabels=le.
↪classes_, yticklabels=le.classes_, ax=axes[1])
        axes[1].set_title(f'Matriz de Confusão para a pior rodada')
        axes[1].set_xlabel('Predito')
        axes[1].set_ylabel('Verdadeiro')
        plt.tight_layout()
        plt.show()

        # OBS 4: Estatísticas por classe (para a melhor rodada)
        print(f'Classification Report for Best Run (#{best_run["run"]}):')
        print(classification_report(best_run['y_test'], best_run['y_pred'],␣
↪target_names=le.classes_))

    return summary_df
```

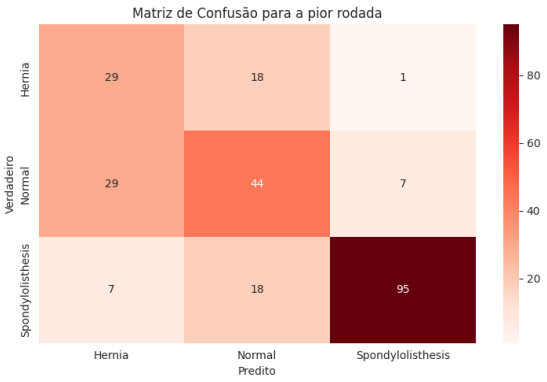
1.0.4 4. Execução Principal e Análise por Split de Teste

[7]:
```
test_splits = [0.8, 0.7, 0.5, 0.3, 0.2] # Corresponde a 20/80, 30/70, 50/50, 70/
↪30, 80/20 train/test
all_split_results = {}

resultados_df = pd.DataFrame()

for split in test_splits:
    train_ratio = int(split*100)
    test_ratio = int((1-split)*100)
    print(f'=================================================================')
    print(f'   Rodando Experimento para Divisão: {train_ratio}/{test_ratio}   ')
    print(f'=================================================================')

    results, le = run_experiments(X, y_raw, test_size=split, n_runs=100)
    resultado = analyze_results(results, le, train_ratio)
```

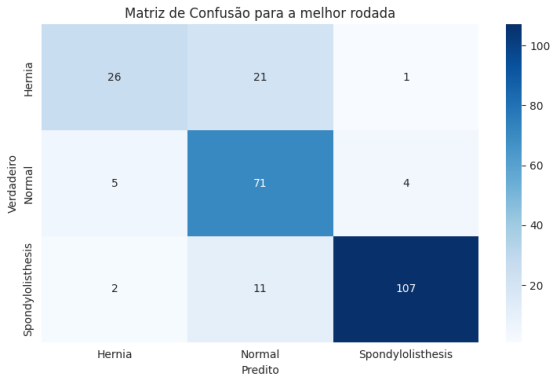
```
    resultados_df = pd.concat([resultados_df, resultado], ignore_index=True)
    all_split_results[f'{train_ratio}/{test_ratio}'] = results
```

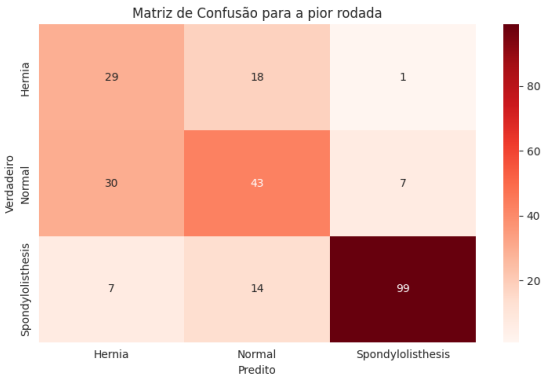
```
====================================================================
   Rodando Experimento para Divisão: 80/19
====================================================================

     Classifier  Acuracia Media  Std Acuracia  Min Acuracia  Max Acuracia  \
4  KNN (m=2.00)        0.784516      0.028124      0.685484      0.834677
3  KNN (m=1.50)        0.784516      0.029709      0.669355      0.842742
5  KNN (m=2.50)        0.783669      0.027309      0.673387      0.834677
2  KNN (m=1.00)        0.778710      0.029134      0.693548      0.846774
7   MDC Robusto        0.772944      0.033970      0.673387      0.846774
1  KNN (m=0.67)        0.765847      0.027213      0.689516      0.822581
6           MDC        0.760565      0.045114      0.592742      0.846774
0  KNN (m=0.50)        0.756815      0.028247      0.677419      0.830645
8         MAXCO        0.713024      0.052021      0.540323      0.802419

   Median Acuracia  Tempo de Treino Medio (ms)  Tempo de Treino Std (ms)  \
4         0.786290                    0.002615                  0.000628
3         0.790323                    0.002701                  0.000761
5         0.786290                    0.002558                  0.000533
2         0.778226                    0.002763                  0.000727
7         0.776210                    0.125999                  0.010985
1         0.768145                    0.002799                  0.000591
6         0.764113                    0.065658                  0.007431
0         0.758065                    0.003850                  0.000868
8         0.719758                    0.049694                  0.004752

   Tempo de Teste Medio (ms)  Tempo de Teste Std (ms)
4                   0.019709                 0.001248
3                   0.025363                 0.002210
5                   0.025053                 0.001216
2                   0.020113                 0.001930
7                   0.005366                 0.000243
1                   0.025439                 0.002218
6                   0.005409                 0.000365
0                   0.020143                 0.001590
8                   0.036468                 0.002619

--- KNN (m=0.50) ---
Melhor rodada (#98) Acurácia: 0.8306
Pior rodada (#77) Acurácia: 0.6774
```





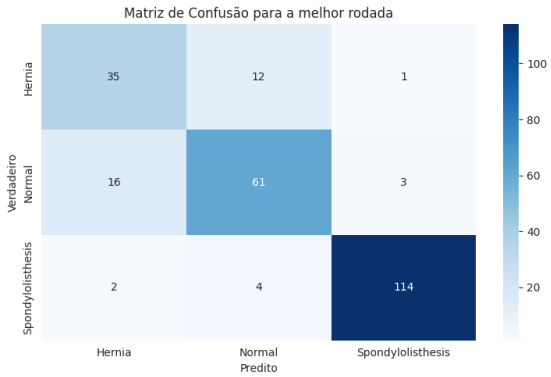

```
Classification Report for Best Run (#98):
                   precision    recall  f1-score   support

           Hernia       0.66      0.69      0.67        48
           Normal       0.78      0.78      0.78        80
Spondylolisthesis       0.93      0.93      0.93       120

         accuracy                           0.83       248
        macro avg       0.79      0.80      0.79       248
     weighted avg       0.83      0.83      0.83       248

--- KNN (m=0.67) ---
Melhor rodada (#0) Acurácia: 0.8226
Pior rodada (#77) Acurácia: 0.6895
```

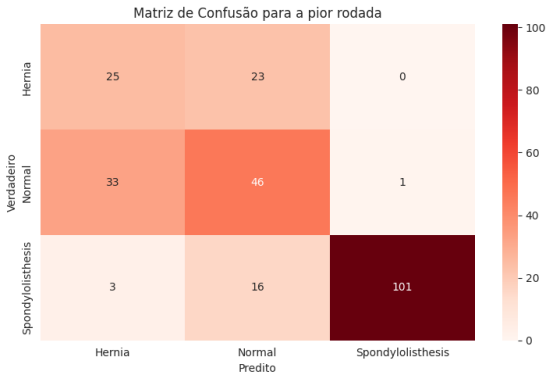




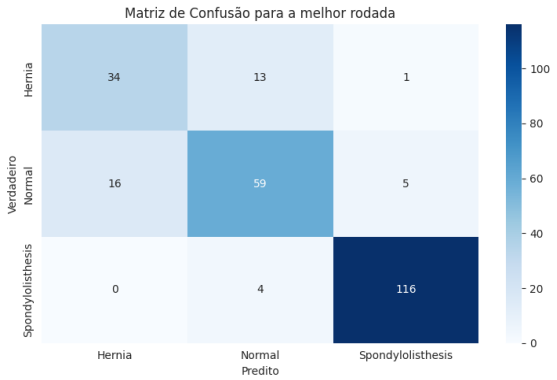

```
Classification Report for Best Run (#0):
                   precision    recall  f1-score   support

           Hernia       0.79      0.54      0.64        48
           Normal       0.69      0.89      0.78        80
```

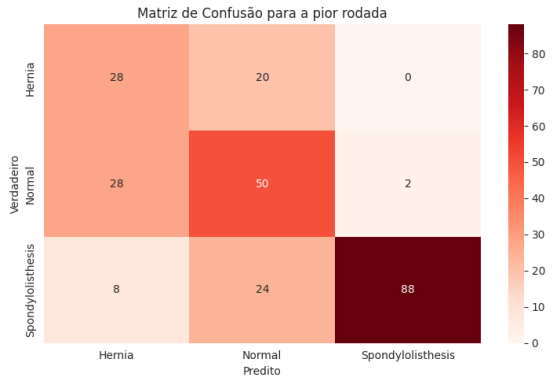
```
Spondylolisthesis       0.96      0.89      0.92       120

         accuracy                           0.82       248
        macro avg       0.81      0.77      0.78       248
     weighted avg       0.84      0.82      0.82       248

--- KNN (m=1.00) ---
Melhor rodada (#98) Acurácia: 0.8468
Pior rodada (#22) Acurácia: 0.6935
```



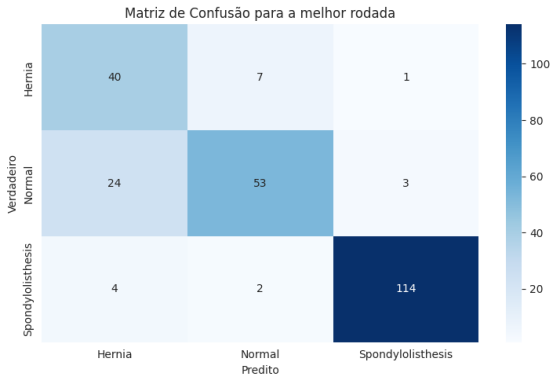


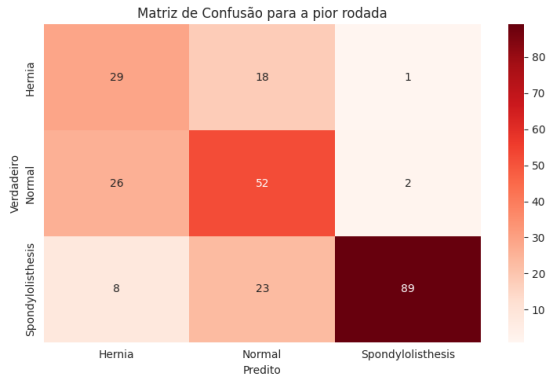

```
Classification Report for Best Run (#98):
                  precision    recall  f1-score   support

           Hernia       0.66      0.73      0.69        48
           Normal       0.79      0.76      0.78        80
Spondylolisthesis       0.97      0.95      0.96       120

         accuracy                           0.85       248
        macro avg       0.81      0.81      0.81       248
     weighted avg       0.85      0.85      0.85       248

--- KNN (m=1.50) ---
Melhor rodada (#1) Acurácia: 0.8427
Pior rodada (#31) Acurácia: 0.6694
```





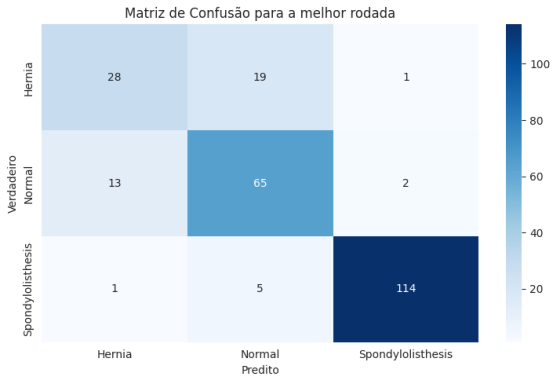

```
Classification Report for Best Run (#1):
                   precision    recall  f1-score   support

           Hernia       0.68      0.71      0.69        48
           Normal       0.78      0.74      0.76        80
Spondylolisthesis       0.95      0.97      0.96       120

         accuracy                           0.84       248
        macro avg       0.80      0.80      0.80       248
     weighted avg       0.84      0.84      0.84       248

--- KNN (m=2.00) ---
Melhor rodada (#87) Acurácia: 0.8347
Pior rodada (#31) Acurácia: 0.6855
```

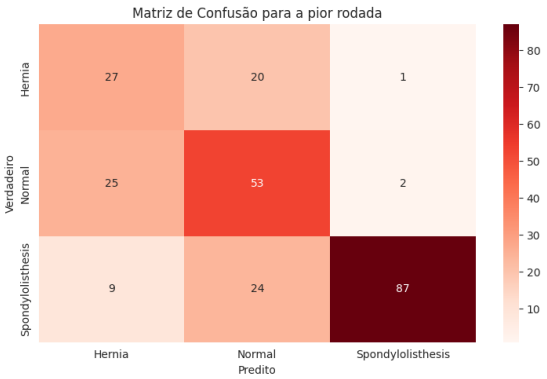




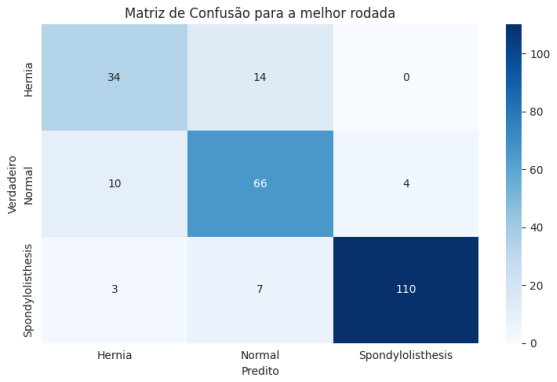

```
Classification Report for Best Run (#87):
                   precision    recall  f1-score   support

           Hernia       0.59      0.83      0.69        48
           Normal       0.85      0.66      0.75        80
```

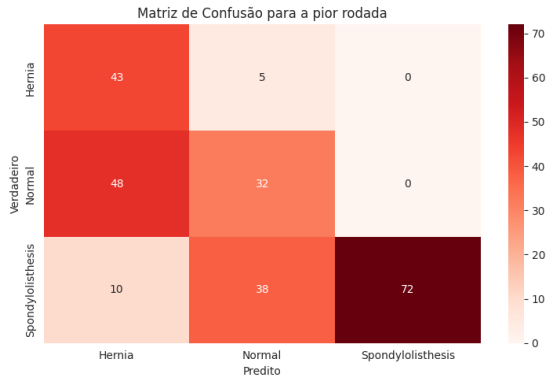
```
Spondylolisthesis       0.97      0.95      0.96       120

         accuracy                           0.83       248
        macro avg       0.80      0.82      0.80       248
     weighted avg       0.86      0.83      0.84       248

--- KNN (m=2.50) ---
Melhor rodada (#6) Acurácia: 0.8347
Pior rodada (#31) Acurácia: 0.6734
```



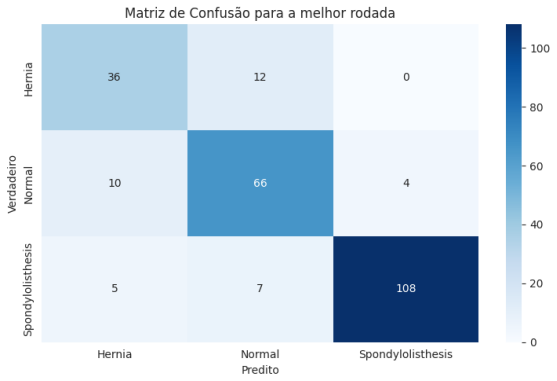


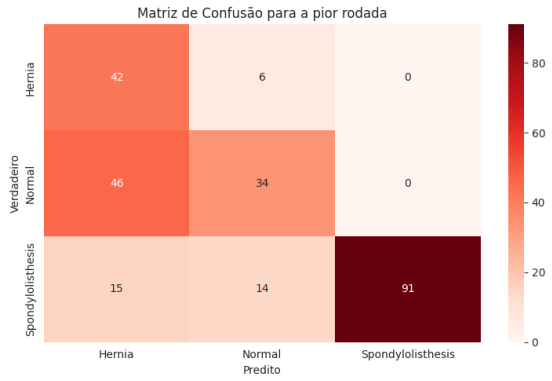

```
Classification Report for Best Run (#6):
                  precision    recall  f1-score   support

           Hernia       0.67      0.58      0.62        48
           Normal       0.73      0.81      0.77        80
Spondylolisthesis       0.97      0.95      0.96       120

         accuracy                           0.83       248
        macro avg       0.79      0.78      0.78       248
     weighted avg       0.84      0.83      0.83       248

--- MDC ---
Melhor rodada (#6) Acurácia: 0.8468
Pior rodada (#31) Acurácia: 0.5927
```





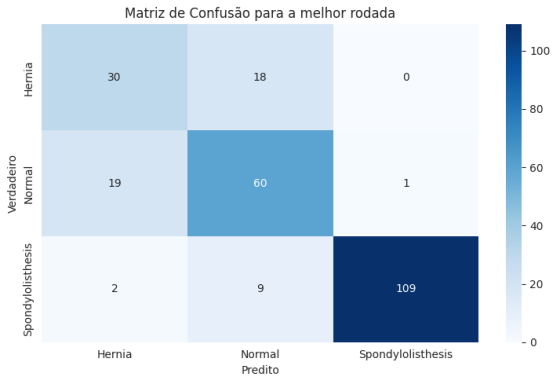

```
Classification Report for Best Run (#6):
                   precision    recall  f1-score   support

           Hernia       0.72      0.71      0.72        48
           Normal       0.76      0.82      0.79        80
Spondylolisthesis       0.96      0.92      0.94       120

         accuracy                           0.85       248
        macro avg       0.82      0.82      0.82       248
     weighted avg       0.85      0.85      0.85       248

--- MDC Robusto ---
Melhor rodada (#83) Acurácia: 0.8468
Pior rodada (#31) Acurácia: 0.6734
```

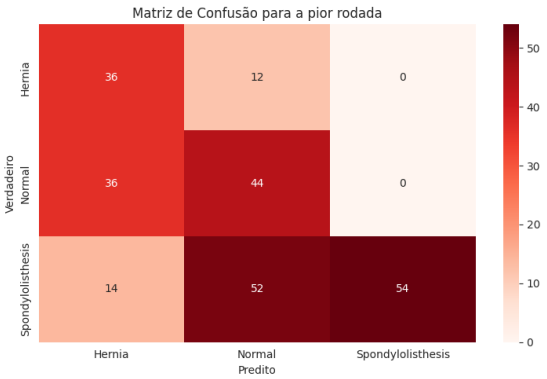





```
Classification Report for Best Run (#83):
                   precision    recall  f1-score   support

           Hernia       0.71      0.75      0.73        48
           Normal       0.78      0.82      0.80        80
```

```
Spondylolisthesis       0.96      0.90      0.93       120

         accuracy                          0.85       248
        macro avg       0.82      0.83      0.82       248
     weighted avg       0.85      0.85      0.85       248

--- MAXCO ---
Melhor rodada (#84) Acurácia: 0.8024
Pior rodada (#76) Acurácia: 0.5403
```

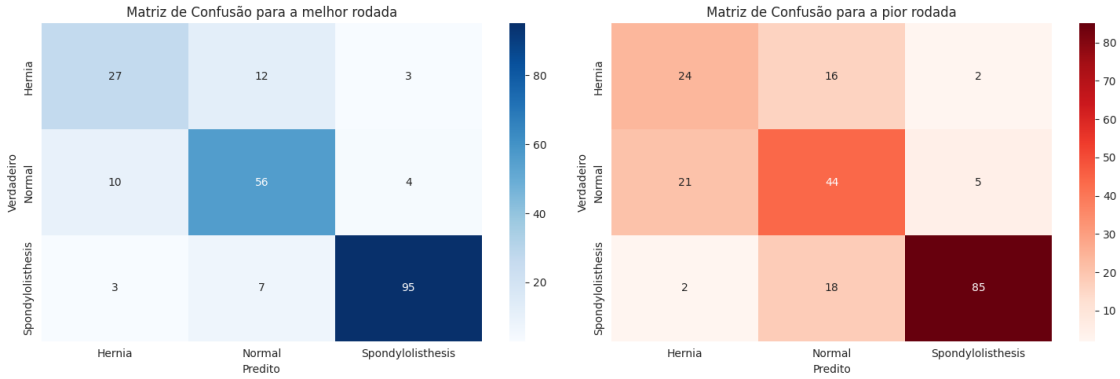





```
Classification Report for Best Run (#84):
                   precision    recall  f1-score   support

           Hernia       0.59      0.62      0.61        48
           Normal       0.69      0.75      0.72        80
Spondylolisthesis       0.99      0.91      0.95       120

         accuracy                          0.80       248
        macro avg       0.76      0.76      0.76       248
     weighted avg       0.82      0.80      0.81       248

================================================================
   Rodando Experimento para Divisão: 70/30
================================================================

     Classifier  Acuracia Media  Std Acuracia  Min Acuracia  Max Acuracia  \
4  KNN (m=2.00)        0.794931      0.028351      0.714286      0.857143
5  KNN (m=2.50)        0.793641      0.027523      0.695853      0.852535
3  KNN (m=1.50)        0.792488      0.030010      0.691244      0.861751
2  KNN (m=1.00)        0.786498      0.027811      0.700461      0.847926
7   MDC Robusto        0.777327      0.029700      0.649770      0.838710
1  KNN (m=0.67)        0.771935      0.024012      0.691244      0.829493
6           MDC        0.765115      0.034837      0.663594      0.843318
0  KNN (m=0.50)        0.760276      0.024393      0.705069      0.820276
8         MAXCO        0.719263      0.043843      0.599078      0.829493
```

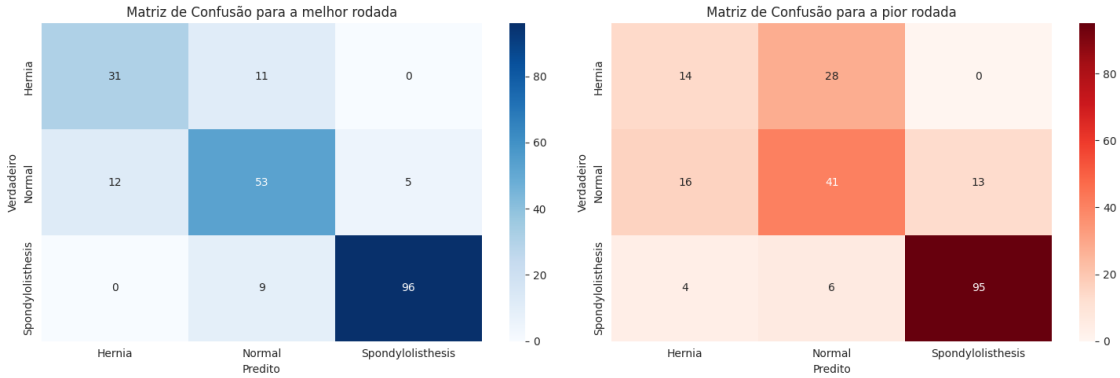
```
   Median Acuracia  Tempo de Treino Medio (ms)  Tempo de Treino Std (ms)  \
4         0.797235                    0.002575                  0.000738
5         0.797235                    0.002468                  0.000388
3         0.792627                    0.002477                  0.000290
2         0.790323                    0.002584                  0.000785
7         0.778802                    0.129135                  0.007447
1         0.774194                    0.002632                  0.000371
6         0.769585                    0.066926                  0.005259
0         0.764977                    0.003626                  0.000542
8         0.728111                    0.050995                  0.003827

   Tempo de Teste Medio (ms)  Tempo de Teste Std (ms)
4                   0.021472                 0.000593
5                   0.029190                 0.000469
3                   0.029229                 0.000441
2                   0.021595                 0.000296
7                   0.005364                 0.000097
1                   0.029279                 0.000478
6                   0.005392                 0.000147
0                   0.021929                 0.000493
8                   0.036215                 0.001371
```

```
--- KNN (m=0.50) ---
Melhor rodada (#98) Acurácia: 0.8203
Pior rodada (#34) Acurácia: 0.7051
```

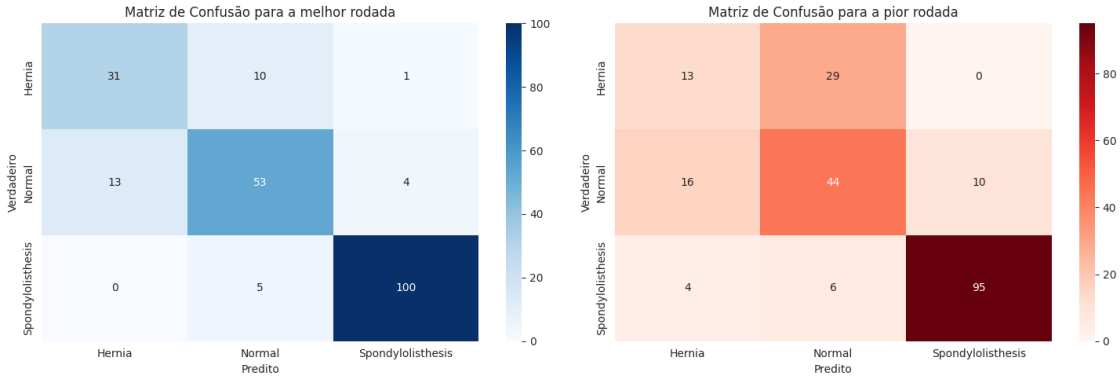



```
Classification Report for Best Run (#98):
                   precision    recall  f1-score   support

           Hernia       0.68      0.64      0.66        42
           Normal       0.75      0.80      0.77        70
Spondylolisthesis       0.93      0.90      0.92       105

         accuracy                           0.82       217
```

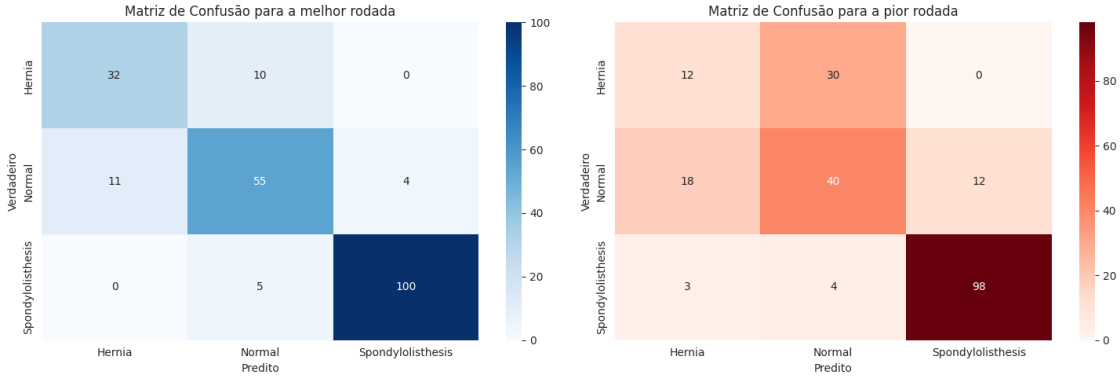
```
       macro avg       0.78      0.78      0.78       217
    weighted avg       0.82      0.82      0.82       217

--- KNN (m=0.67) ---
Melhor rodada (#65) Acurácia: 0.8295
Pior rodada (#49) Acurácia: 0.6912
```



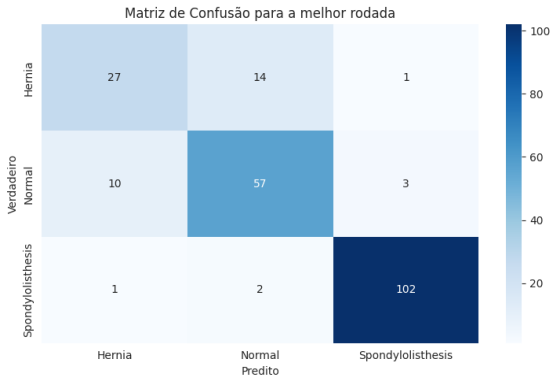

```
Classification Report for Best Run (#65):
                   precision    recall  f1-score   support

           Hernia       0.72      0.74      0.73        42
           Normal       0.73      0.76      0.74        70
Spondylolisthesis       0.95      0.91      0.93       105

         accuracy                           0.83       217
        macro avg       0.80      0.80      0.80       217
     weighted avg       0.83      0.83      0.83       217

--- KNN (m=1.00) ---
Melhor rodada (#1) Acurácia: 0.8479
Pior rodada (#49) Acurácia: 0.7005
```

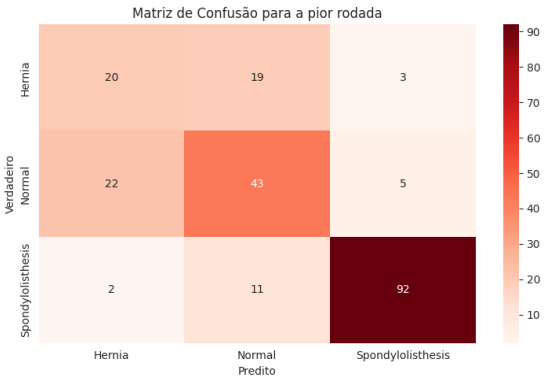


```
Classification Report for Best Run (#1):
                   precision    recall  f1-score   support

           Hernia       0.70      0.74      0.72        42
           Normal       0.78      0.76      0.77        70
Spondylolisthesis       0.95      0.95      0.95       105

         accuracy                           0.85       217
        macro avg       0.81      0.82      0.81       217
     weighted avg       0.85      0.85      0.85       217

--- KNN (m=1.50) ---
Melhor rodada (#1) Acurácia: 0.8618
Pior rodada (#49) Acurácia: 0.6912
```

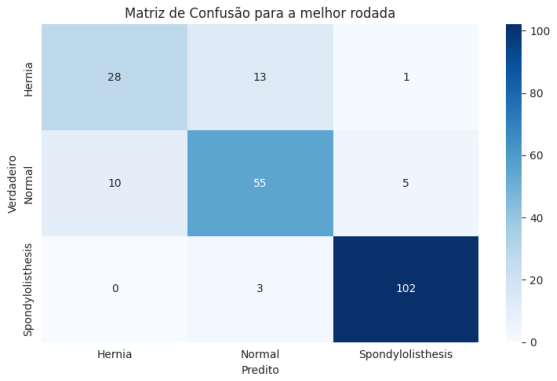


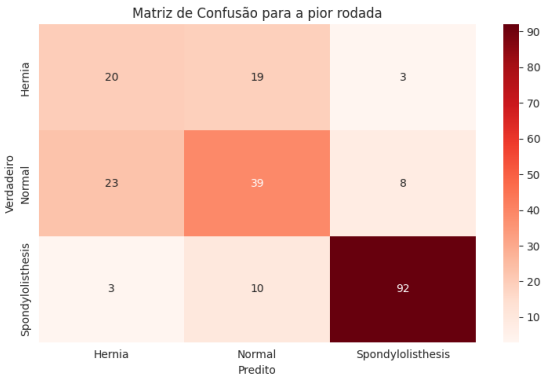

```
Classification Report for Best Run (#1):
                   precision    recall  f1-score   support

           Hernia       0.74      0.76      0.75        42
           Normal       0.79      0.79      0.79        70
Spondylolisthesis       0.96      0.95      0.96       105

         accuracy                           0.86       217
        macro avg       0.83      0.83      0.83       217
     weighted avg       0.86      0.86      0.86       217

--- KNN (m=2.00) ---
Melhor rodada (#6) Acurácia: 0.8571
Pior rodada (#42) Acurácia: 0.7143
```





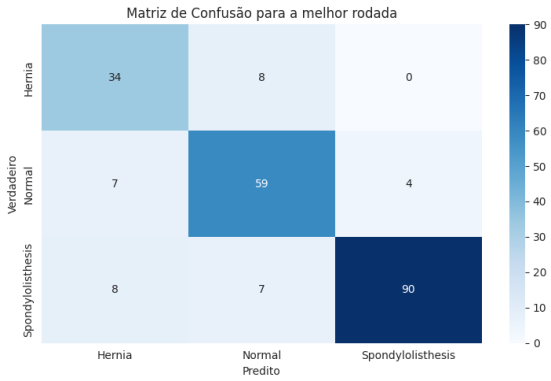

```
Classification Report for Best Run (#6):
                   precision    recall  f1-score   support

           Hernia       0.71      0.64      0.68        42
           Normal       0.78      0.81      0.80        70
Spondylolisthesis       0.96      0.97      0.97       105

         accuracy                           0.86       217
        macro avg       0.82      0.81      0.81       217
     weighted avg       0.86      0.86      0.86       217

--- KNN (m=2.50) ---
Melhor rodada (#8) Acurácia: 0.8525
Pior rodada (#42) Acurácia: 0.6959
```

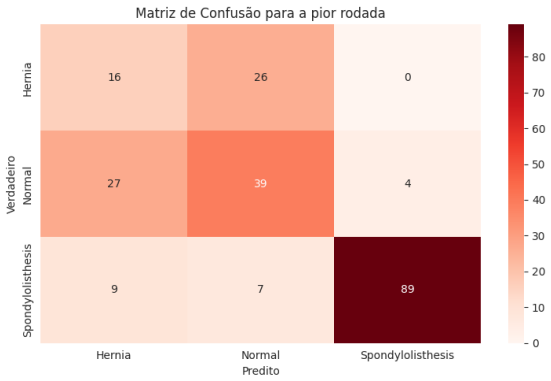




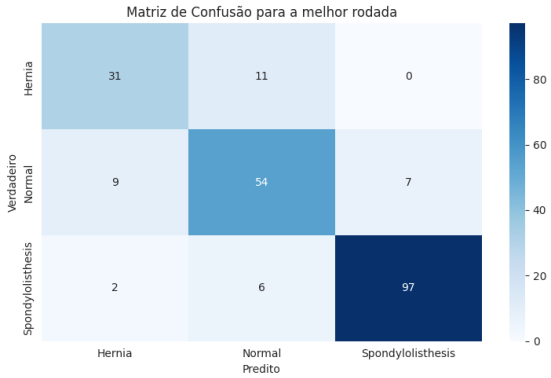

```
Classification Report for Best Run (#8):
                   precision    recall  f1-score   support

           Hernia       0.74      0.67      0.70        42
           Normal       0.77      0.79      0.78        70
```

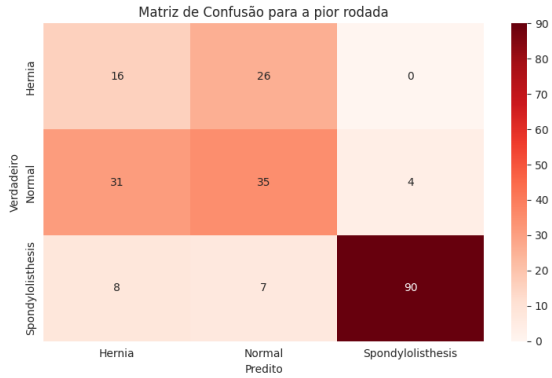
```
Spondylolisthesis       0.94      0.97      0.96       105

         accuracy                           0.85       217
        macro avg       0.82      0.81      0.81       217
     weighted avg       0.85      0.85      0.85       217

--- MDC ---
Melhor rodada (#59) Acurácia: 0.8433
Pior rodada (#49) Acurácia: 0.6636
```



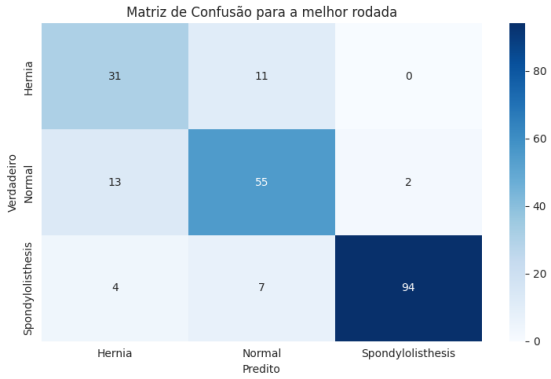


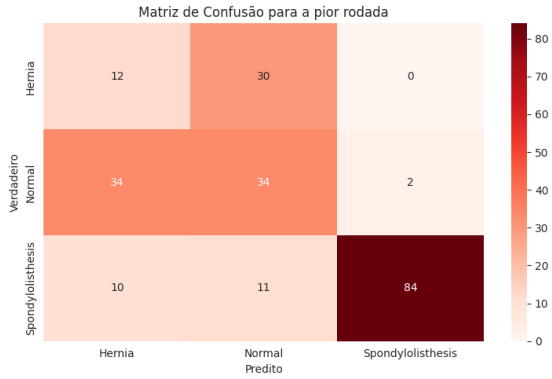

```
Classification Report for Best Run (#59):
                  precision    recall  f1-score   support

           Hernia       0.69      0.81      0.75        42
           Normal       0.80      0.84      0.82        70
Spondylolisthesis       0.96      0.86      0.90       105

         accuracy                           0.84       217
        macro avg       0.82      0.84      0.82       217
     weighted avg       0.85      0.84      0.85       217

--- MDC Robusto ---
Melhor rodada (#6) Acurácia: 0.8387
Pior rodada (#49) Acurácia: 0.6498
```





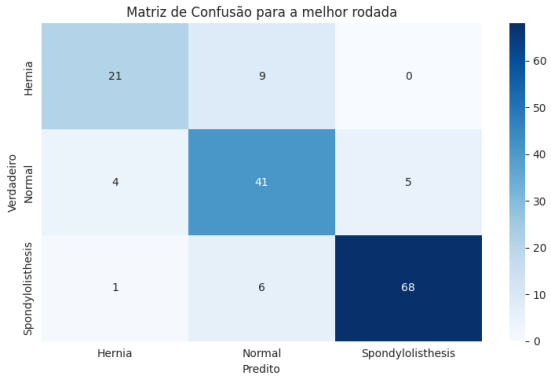

```
Classification Report for Best Run (#6):
                   precision    recall  f1-score   support

           Hernia       0.74      0.74      0.74        42
           Normal       0.76      0.77      0.77        70
Spondylolisthesis       0.93      0.92      0.93       105

         accuracy                           0.84       217
        macro avg       0.81      0.81      0.81       217
     weighted avg       0.84      0.84      0.84       217

--- MAXCO ---
Melhor rodada (#69) Acurácia: 0.8295
Pior rodada (#49) Acurácia: 0.5991
```

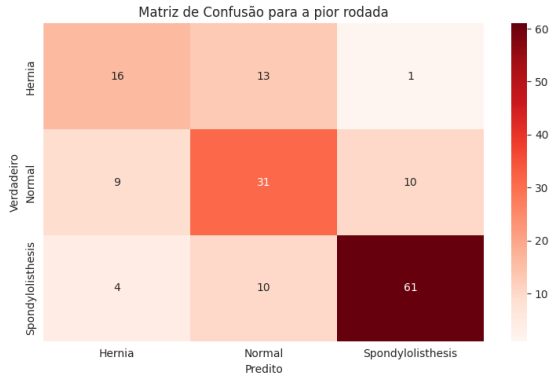




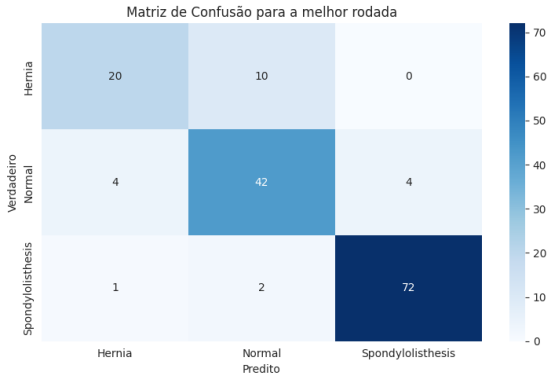

```
Classification Report for Best Run (#69):
                   precision    recall  f1-score   support

           Hernia       0.65      0.74      0.69        42
           Normal       0.75      0.79      0.77        70
```

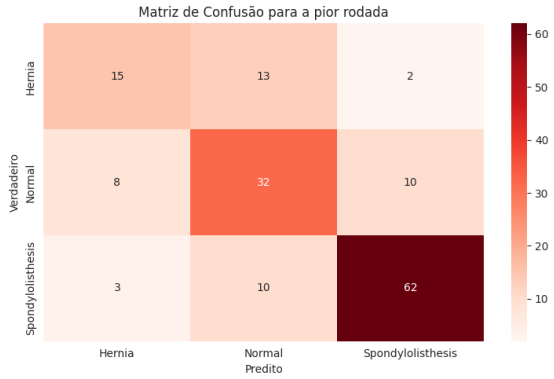
```
Spondylolisthesis       0.98      0.90      0.94       105

         accuracy                           0.83       217
        macro avg       0.79      0.81      0.80       217
     weighted avg       0.84      0.83      0.83       217

==================================================================
   Rodando Experimento para Divisão: 50/50
==================================================================

     Classifier  Acuracia Media  Std Acuracia  Min Acuracia  Max Acuracia  \
4  KNN (m=2.00)        0.806581      0.027265      0.741935      0.890323
5  KNN (m=2.50)        0.806194      0.026880      0.741935      0.883871
3  KNN (m=1.50)        0.804065      0.027834      0.754839      0.890323
2  KNN (m=1.00)        0.797677      0.029403      0.722581      0.877419
7   MDC Robusto        0.785548      0.028840      0.696774      0.851613
1  KNN (m=0.67)        0.779613      0.027169      0.703226      0.864516
6           MDC        0.770645      0.033751      0.677419      0.851613
0  KNN (m=0.50)        0.766774      0.028098      0.696774      0.838710
8         MAXCO        0.723484      0.036610      0.619355      0.800000

   Median Acuracia  Tempo de Treino Medio (ms)  Tempo de Treino Std (ms)  \
4         0.806452                    0.002403                  0.000407
5         0.806452                    0.002384                  0.000380
3         0.806452                    0.002434                  0.000329
2         0.796774                    0.002503                  0.000594
7         0.787097                    0.132589                  0.006075
1         0.780645                    0.002525                  0.000296
6         0.774194                    0.067036                  0.004947
0         0.767742                    0.003555                  0.000506
8         0.722581                    0.053742                  0.006205

   Tempo de Teste Medio (ms)  Tempo de Teste Std (ms)
4                   0.025554                 0.000753
5                   0.038123                 0.000621
3                   0.038218                 0.001106
2                   0.025748                 0.000827
7                   0.005476                 0.000139
1                   0.038298                 0.001188
6                   0.005491                 0.000193
0                   0.026328                 0.000596
8                   0.036694                 0.001032

--- KNN (m=0.50) ---
Melhor rodada (#47) Acurácia: 0.8387
Pior rodada (#59) Acurácia: 0.6968
```





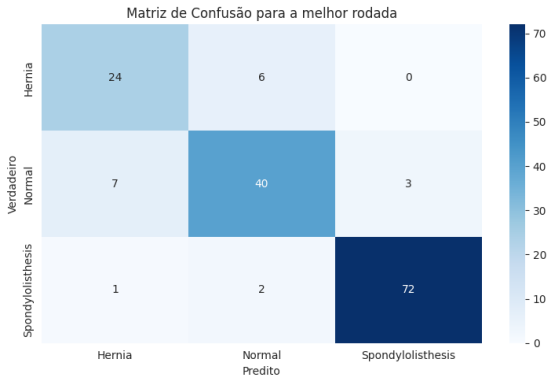

```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.81      0.70      0.75        30
           Normal       0.73      0.82      0.77        50
Spondylolisthesis       0.93      0.91      0.92        75

         accuracy                           0.84       155
        macro avg       0.82      0.81      0.81       155
     weighted avg       0.84      0.84      0.84       155

--- KNN (m=0.67) ---
Melhor rodada (#47) Acurácia: 0.8645
Pior rodada (#59) Acurácia: 0.7032
```

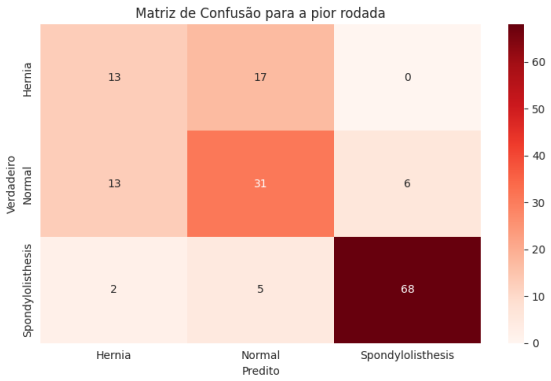




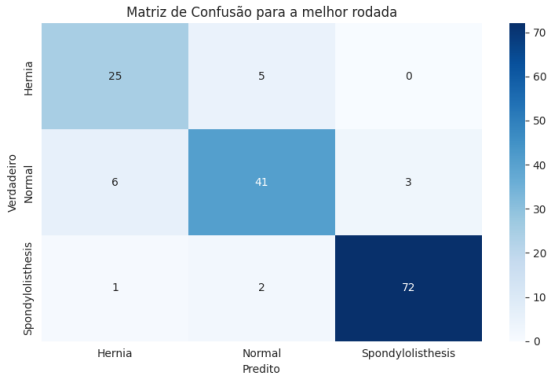

```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.80      0.67      0.73        30
           Normal       0.78      0.84      0.81        50
```

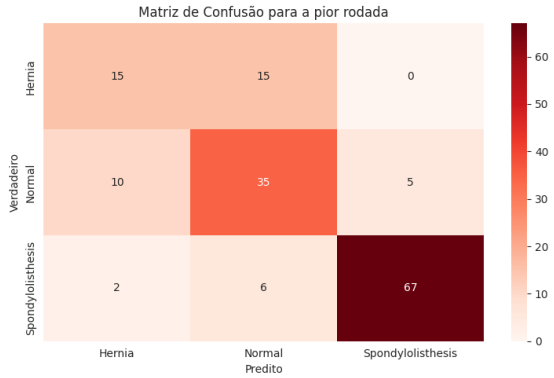
```
Spondylolisthesis       0.95      0.96      0.95        75

         accuracy                          0.86       155
        macro avg       0.84      0.82      0.83       155
     weighted avg       0.86      0.86      0.86       155

--- KNN (m=1.00) ---
Melhor rodada (#56) Acurácia: 0.8774
Pior rodada (#43) Acurácia: 0.7226
```



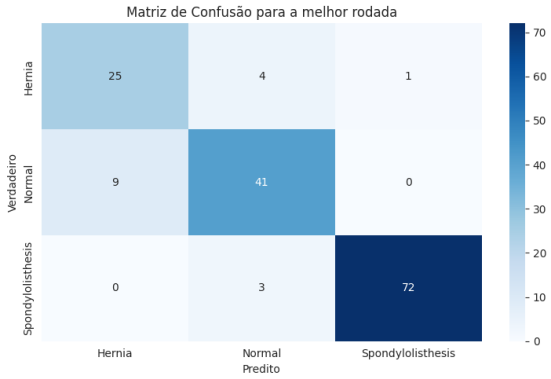


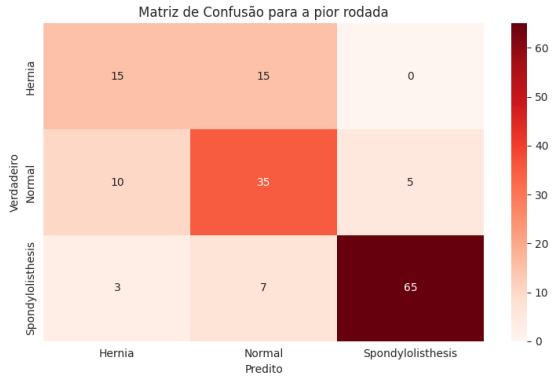

```
Classification Report for Best Run (#56):
                   precision    recall  f1-score   support

           Hernia       0.75      0.80      0.77        30
           Normal       0.83      0.80      0.82        50
Spondylolisthesis       0.96      0.96      0.96        75

         accuracy                          0.88       155
        macro avg       0.85      0.85      0.85       155
     weighted avg       0.88      0.88      0.88       155

--- KNN (m=1.50) ---
Melhor rodada (#56) Acurácia: 0.8903
Pior rodada (#11) Acurácia: 0.7548
```





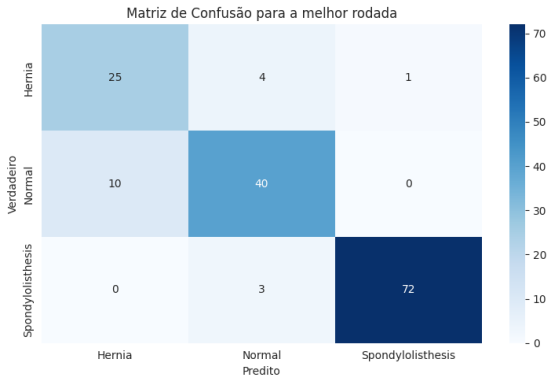

```
Classification Report for Best Run (#56):
                   precision    recall  f1-score   support

           Hernia       0.78      0.83      0.81        30
           Normal       0.85      0.82      0.84        50
Spondylolisthesis       0.96      0.96      0.96        75

         accuracy                           0.89       155
        macro avg       0.87      0.87      0.87       155
     weighted avg       0.89      0.89      0.89       155

--- KNN (m=2.00) ---
Melhor rodada (#4) Acurácia: 0.8903
Pior rodada (#11) Acurácia: 0.7419
```

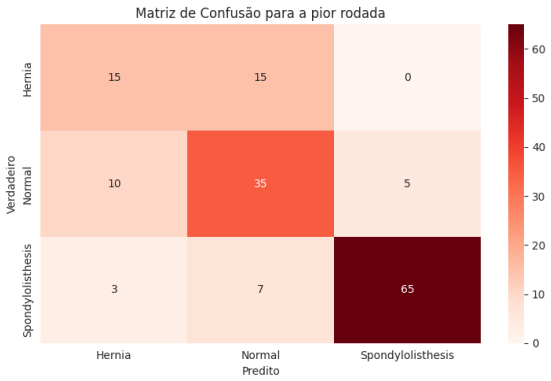




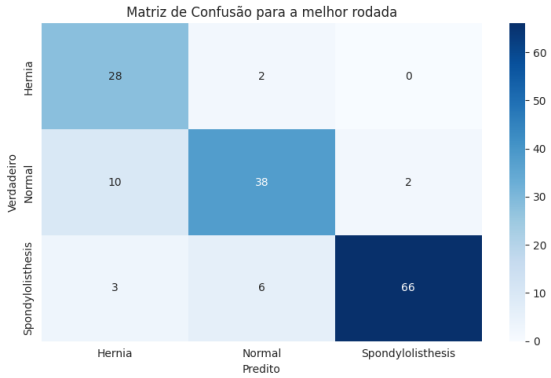

```
Classification Report for Best Run (#4):
                   precision    recall  f1-score   support

           Hernia       0.74      0.83      0.78        30
           Normal       0.85      0.82      0.84        50
```

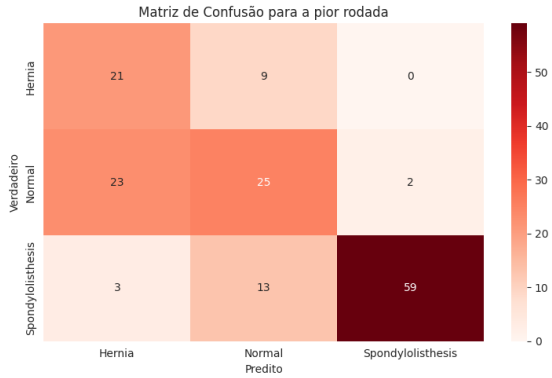
```
Spondylolisthesis       0.99      0.96      0.97        75

         accuracy                           0.89       155
        macro avg       0.86      0.87      0.86       155
     weighted avg       0.90      0.89      0.89       155

--- KNN (m=2.50) ---
Melhor rodada (#4) Acurácia: 0.8839
Pior rodada (#11) Acurácia: 0.7419
```



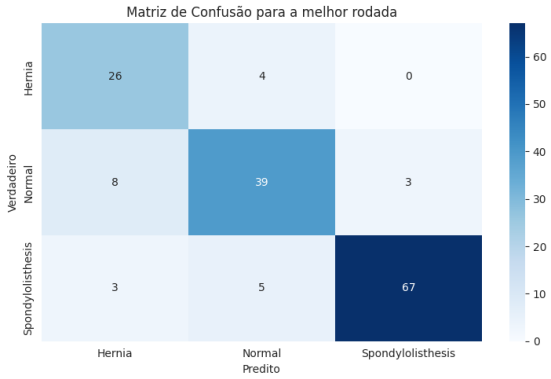


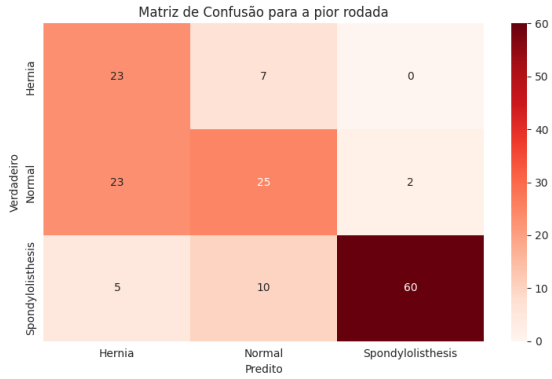

```
Classification Report for Best Run (#4):
                  precision    recall  f1-score   support

           Hernia       0.71      0.83      0.77        30
           Normal       0.85      0.80      0.82        50
Spondylolisthesis       0.99      0.96      0.97        75

         accuracy                           0.88       155
        macro avg       0.85      0.86      0.86       155
     weighted avg       0.89      0.88      0.89       155

--- MDC ---
Melhor rodada (#24) Acurácia: 0.8516
Pior rodada (#94) Acurácia: 0.6774
```





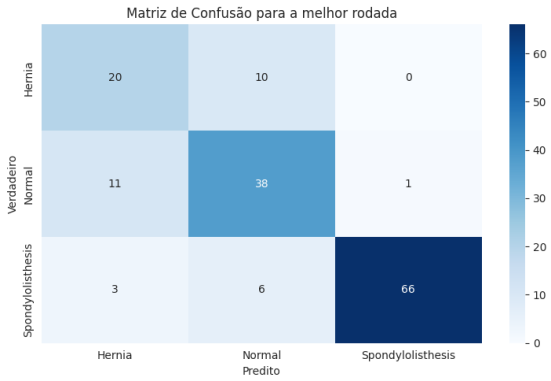

```
Classification Report for Best Run (#24):
                   precision    recall  f1-score   support

           Hernia       0.68      0.93      0.79        30
           Normal       0.83      0.76      0.79        50
Spondylolisthesis       0.97      0.88      0.92        75

         accuracy                           0.85       155
        macro avg       0.83      0.86      0.83       155
     weighted avg       0.87      0.85      0.85       155

--- MDC Robusto ---
Melhor rodada (#56) Acurácia: 0.8516
Pior rodada (#54) Acurácia: 0.6968
```

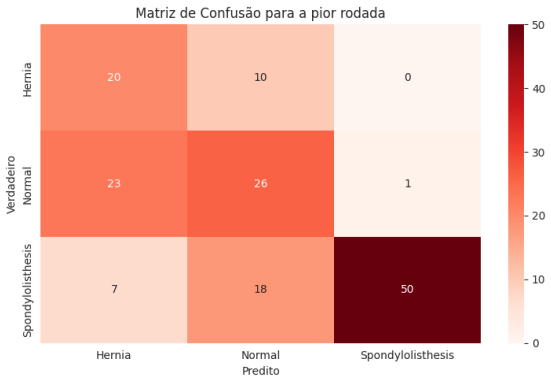





```
Classification Report for Best Run (#56):
                   precision    recall  f1-score   support

           Hernia       0.70      0.87      0.78        30
           Normal       0.81      0.78      0.80        50
```

```
Spondylolisthesis       0.96      0.89      0.92        75

         accuracy                          0.85       155
        macro avg       0.82      0.85      0.83       155
     weighted avg       0.86      0.85      0.85       155

--- MAXCO ---
Melhor rodada (#83) Acurácia: 0.8000
Pior rodada (#54) Acurácia: 0.6194
```

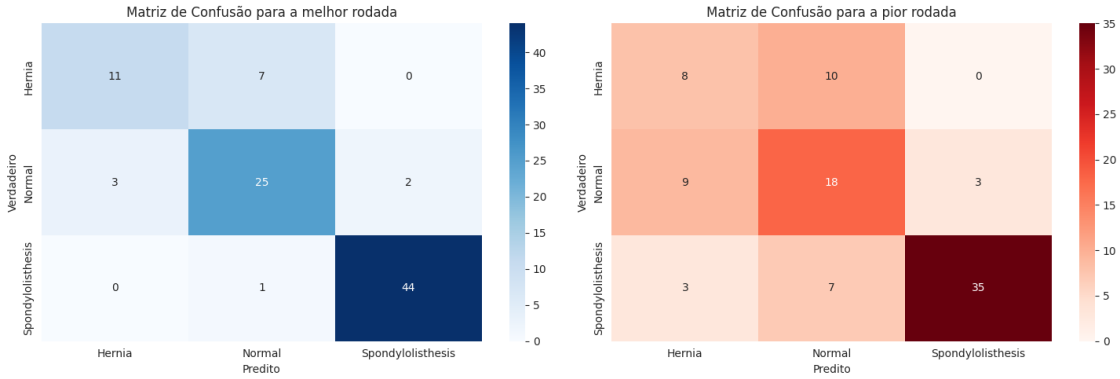





```
Classification Report for Best Run (#83):
                   precision    recall  f1-score   support

           Hernia       0.59      0.67      0.62        30
           Normal       0.70      0.76      0.73        50
Spondylolisthesis       0.99      0.88      0.93        75

         accuracy                          0.80       155
        macro avg       0.76      0.77      0.76       155
     weighted avg       0.82      0.80      0.81       155

================================================================
   Rodando Experimento para Divisão: 30/70
================================================================

     Classifier  Acuracia Media  Std Acuracia  Min Acuracia  Max Acuracia  \
5  KNN (m=2.50)        0.814409      0.034463      0.752688      0.903226
4  KNN (m=2.00)        0.812151      0.035352      0.741935      0.892473
3  KNN (m=1.50)        0.807849      0.033923      0.731183      0.903226
2  KNN (m=1.00)        0.805699      0.037473      0.709677      0.913978
7   MDC Robusto        0.786667      0.037364      0.698925      0.870968
1  KNN (m=0.67)        0.780860      0.037918      0.655914      0.870968
6           MDC        0.765914      0.039679      0.655914      0.870968
0  KNN (m=0.50)        0.765699      0.038927      0.655914      0.860215
8         MAXCO        0.722043      0.046158      0.580645      0.849462
```

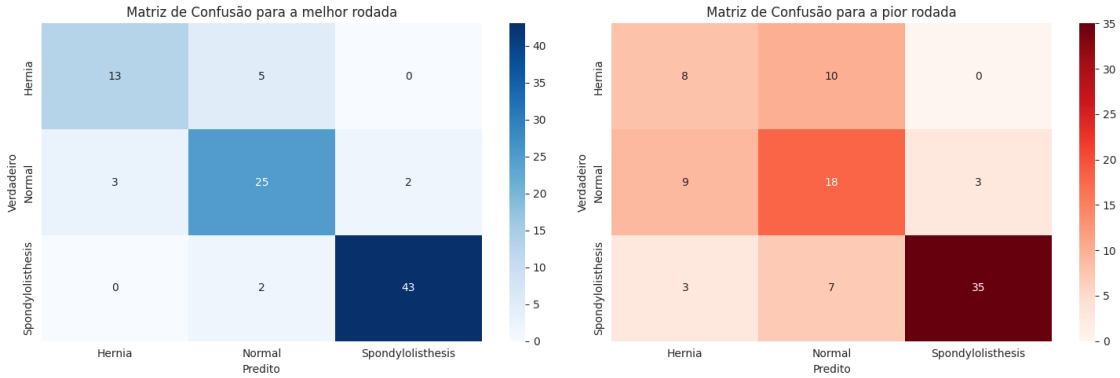
```
   Median Acuracia  Tempo de Treino Medio (ms)  Tempo de Treino Std (ms)  \
5         0.806452                    0.002437                  0.000718
4         0.806452                    0.002480                  0.000777
3         0.806452                    0.002387                  0.000777
2         0.806452                    0.002475                  0.000347
7         0.795699                    0.139074                  0.010912
1         0.784946                    0.002508                  0.000240
6         0.763441                    0.070791                  0.008751
0         0.774194                    0.003502                  0.000435
8         0.720430                    0.055487                  0.003100

   Tempo de Teste Medio (ms)  Tempo de Teste Std (ms)
5                   0.045346                 0.001267
4                   0.027775                 0.001308
3                   0.045197                 0.000940
2                   0.027919                 0.001259
7                   0.005531                 0.000443
1                   0.045318                 0.001132
6                   0.005554                 0.000502
0                   0.028840                 0.000602
8                   0.036736                 0.001013
```

```
--- KNN (m=0.50) ---
Melhor rodada (#98) Acurácia: 0.8602
Pior rodada (#57) Acurácia: 0.6559
```

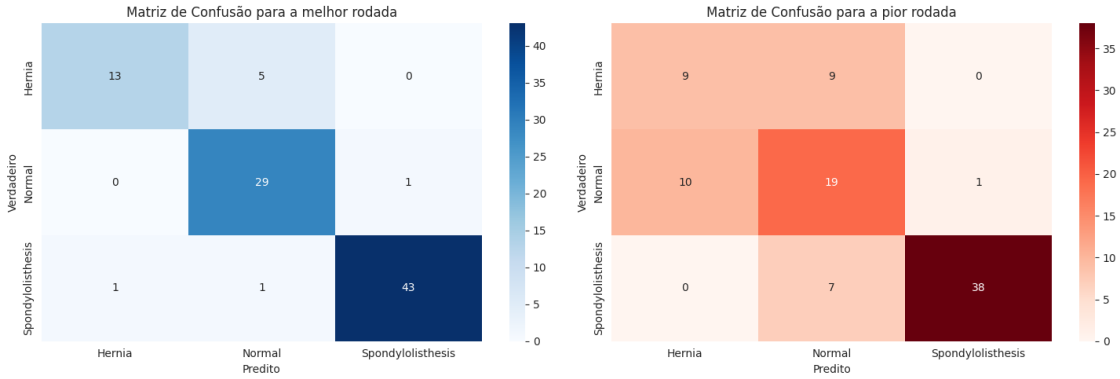



```
Classification Report for Best Run (#98):
                   precision    recall  f1-score   support

           Hernia       0.79      0.61      0.69        18
           Normal       0.76      0.83      0.79        30
Spondylolisthesis       0.96      0.98      0.97        45

         accuracy                           0.86        93
```

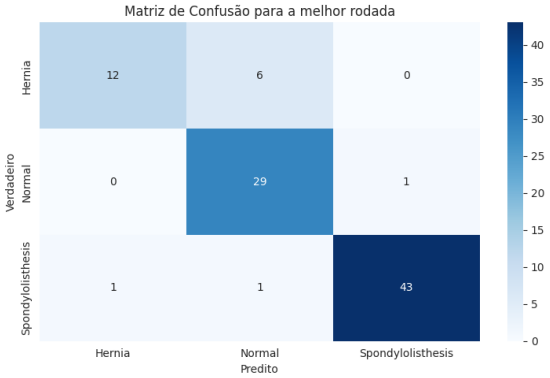
```
        macro avg       0.83      0.81      0.82        93
     weighted avg       0.86      0.86      0.86        93

--- KNN (m=0.67) ---
Melhor rodada (#47) Acurácia: 0.8710
Pior rodada (#57) Acurácia: 0.6559
```

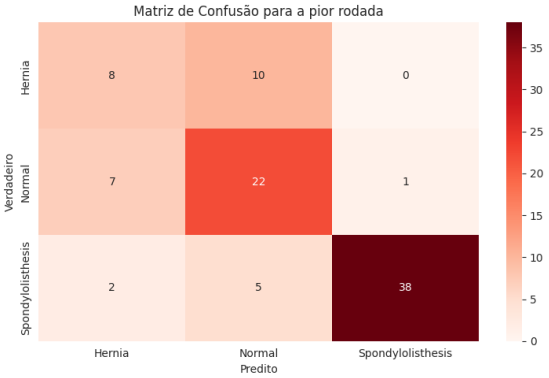



```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.81      0.72      0.76        18
           Normal       0.78      0.83      0.81        30
Spondylolisthesis       0.96      0.96      0.96        45

         accuracy                           0.87        93
        macro avg       0.85      0.84      0.84        93
     weighted avg       0.87      0.87      0.87        93

--- KNN (m=1.00) ---
Melhor rodada (#65) Acurácia: 0.9140
Pior rodada (#12) Acurácia: 0.7097
```

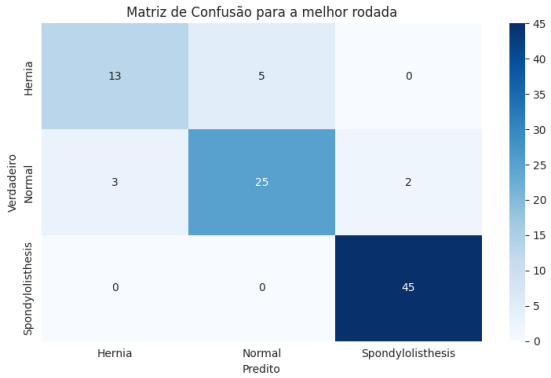


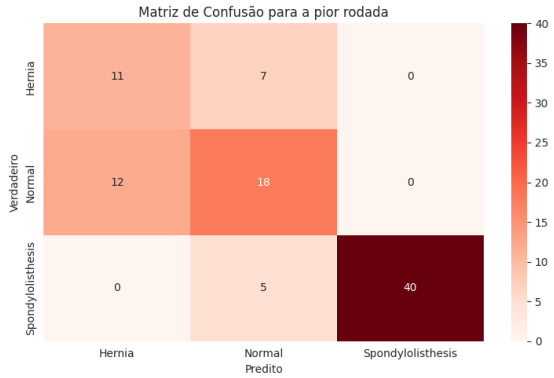
```
Classification Report for Best Run (#65):
                   precision    recall  f1-score   support

           Hernia       0.93      0.72      0.81        18
           Normal       0.83      0.97      0.89        30
Spondylolisthesis       0.98      0.96      0.97        45

         accuracy                           0.91        93
        macro avg       0.91      0.88      0.89        93
     weighted avg       0.92      0.91      0.91        93

--- KNN (m=1.50) ---
Melhor rodada (#65) Acurácia: 0.9032
Pior rodada (#29) Acurácia: 0.7312
```



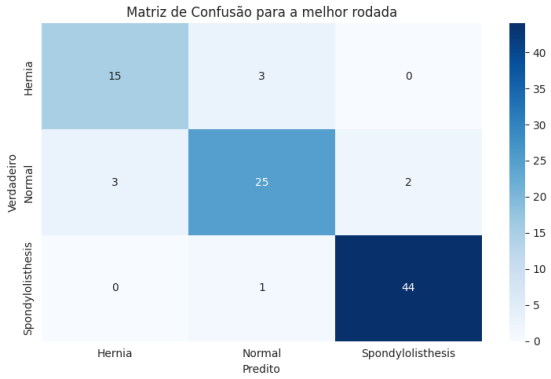


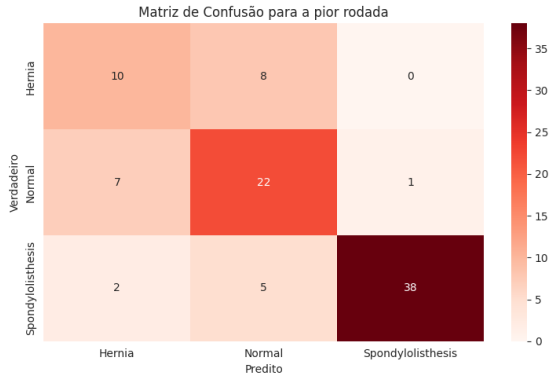

```
Classification Report for Best Run (#65):
                   precision    recall  f1-score   support

           Hernia       0.92      0.67      0.77        18
           Normal       0.81      0.97      0.88        30
Spondylolisthesis       0.98      0.96      0.97        45

         accuracy                           0.90        93
        macro avg       0.90      0.86      0.87        93
     weighted avg       0.91      0.90      0.90        93

--- KNN (m=2.00) ---
Melhor rodada (#23) Acurácia: 0.8925
Pior rodada (#3) Acurácia: 0.7419
```





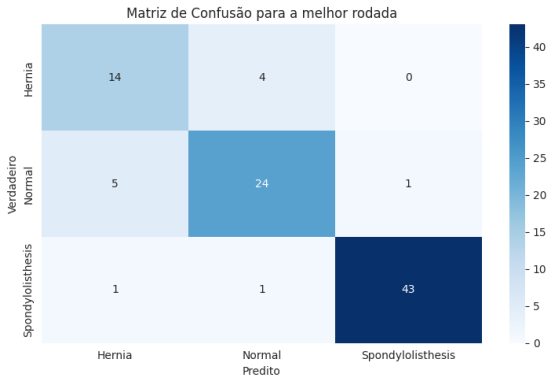

```
Classification Report for Best Run (#23):
                   precision    recall  f1-score   support

           Hernia       0.81      0.72      0.76        18
           Normal       0.83      0.83      0.83        30
Spondylolisthesis       0.96      1.00      0.98        45

         accuracy                           0.89        93
        macro avg       0.87      0.85      0.86        93
     weighted avg       0.89      0.89      0.89        93

--- KNN (m=2.50) ---
Melhor rodada (#99) Acurácia: 0.9032
Pior rodada (#12) Acurácia: 0.7527
```

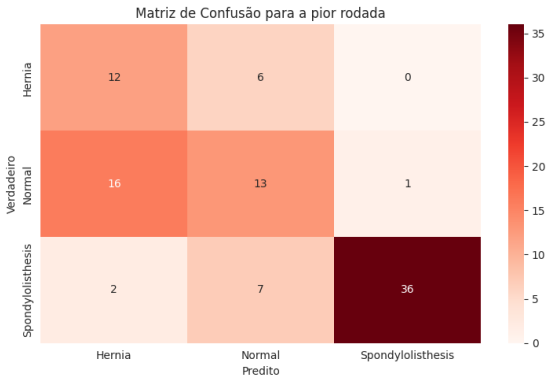




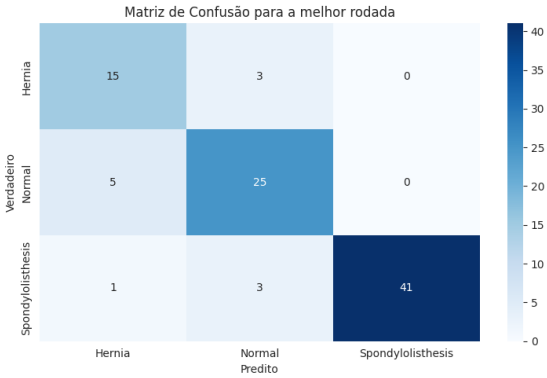

```
Classification Report for Best Run (#99):
                   precision    recall  f1-score   support

           Hernia       0.83      0.83      0.83        18
           Normal       0.86      0.83      0.85        30
```

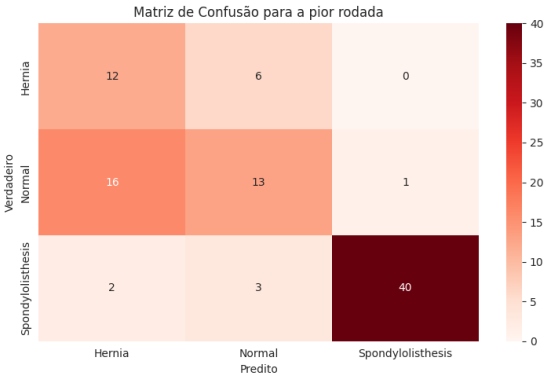
```
Spondylolisthesis       0.96      0.98      0.97        45

         accuracy                           0.90        93
        macro avg       0.88      0.88      0.88        93
     weighted avg       0.90      0.90      0.90        93

--- MDC ---
Melhor rodada (#28) Acurácia: 0.8710
Pior rodada (#3) Acurácia: 0.6559
```



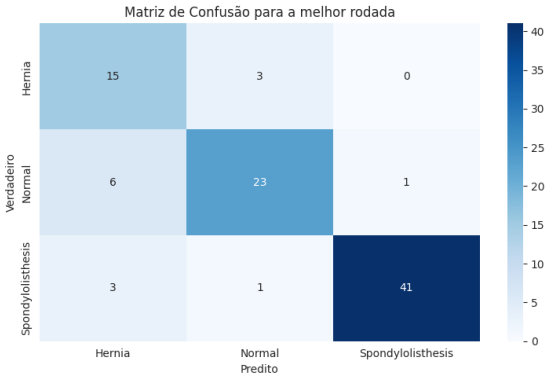


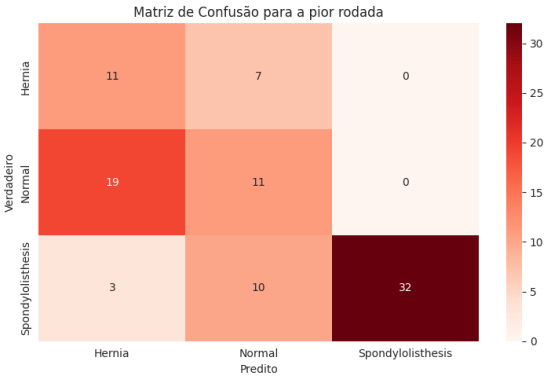

```
Classification Report for Best Run (#28):
                  precision    recall  f1-score   support

           Hernia       0.70      0.78      0.74        18
           Normal       0.83      0.80      0.81        30
Spondylolisthesis       0.98      0.96      0.97        45

         accuracy                           0.87        93
        macro avg       0.83      0.84      0.84        93
     weighted avg       0.88      0.87      0.87        93

--- MDC Robusto ---
Melhor rodada (#80) Acurácia: 0.8710
Pior rodada (#3) Acurácia: 0.6989
```





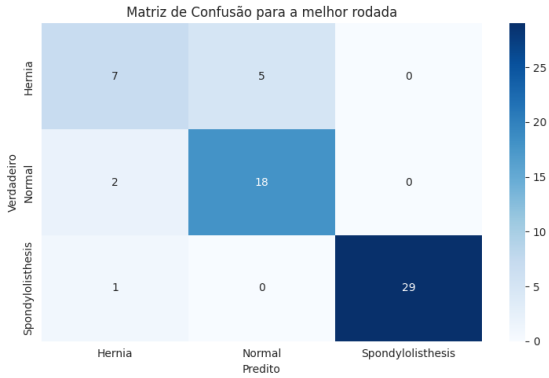

```
Classification Report for Best Run (#80):
                   precision    recall  f1-score   support

           Hernia       0.71      0.83      0.77        18
           Normal       0.81      0.83      0.82        30
Spondylolisthesis       1.00      0.91      0.95        45

         accuracy                           0.87        93
        macro avg       0.84      0.86      0.85        93
     weighted avg       0.88      0.87      0.87        93

--- MAXCO ---
Melhor rodada (#79) Acurácia: 0.8495
Pior rodada (#22) Acurácia: 0.5806
```

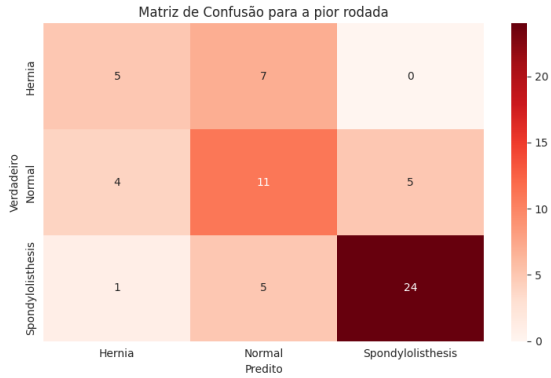





```
Classification Report for Best Run (#79):
                   precision    recall  f1-score   support

           Hernia       0.62      0.83      0.71        18
           Normal       0.85      0.77      0.81        30
```

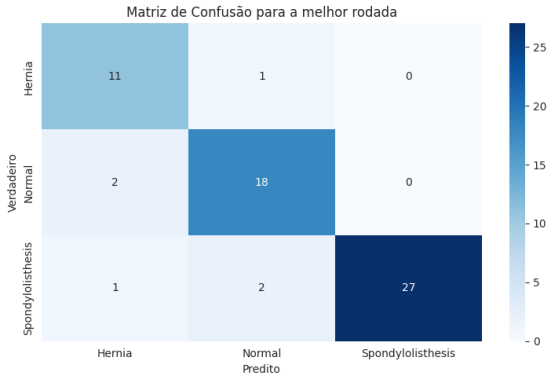
```
Spondylolisthesis       0.98      0.91      0.94        45

         accuracy                           0.85        93
        macro avg       0.82      0.84      0.82        93
     weighted avg       0.87      0.85      0.85        93

==================================================================
   Rodando Experimento para Divisão: 20/80
==================================================================

     Classifier  Acuracia Media  Std Acuracia  Min Acuracia  Max Acuracia  \
5  KNN (m=2.50)        0.823871      0.042040      0.725806      0.919355
4  KNN (m=2.00)        0.820968      0.042275      0.693548      0.903226
3  KNN (m=1.50)        0.816129      0.041624      0.725806      0.903226
2  KNN (m=1.00)        0.815806      0.041196      0.709677      0.935484
7   MDC Robusto        0.786129      0.045024      0.677419      0.870968
1  KNN (m=0.67)        0.784516      0.048949      0.629032      0.903226
0  KNN (m=0.50)        0.765484      0.045243      0.645161      0.870968
6           MDC        0.762903      0.046495      0.661290      0.887097
8         MAXCO        0.722097      0.060084      0.564516      0.887097

   Median Acuracia  Tempo de Treino Medio (ms)  Tempo de Treino Std (ms)  \
5         0.822581                    0.002391                  0.000269
4         0.822581                    0.002372                  0.000266
3         0.806452                    0.002348                  0.000244
2         0.814516                    0.002530                  0.000783
7         0.782258                    0.139830                  0.014982
1         0.790323                    0.002534                  0.000648
0         0.758065                    0.003431                  0.000747
6         0.774194                    0.068069                  0.006299
8         0.709677                    0.057039                  0.004860

   Tempo de Teste Medio (ms)  Tempo de Teste Std (ms)
5                   0.049257                 0.002637
4                   0.029628                 0.002188
3                   0.049399                 0.002059
2                   0.029088                 0.001402
7                   0.005553                 0.000334
1                   0.049910                 0.002270
0                   0.030506                 0.001628
6                   0.005747                 0.000801
8                   0.037629                 0.002178

--- KNN (m=0.50) ---
Melhor rodada (#92) Acurácia: 0.8710
Pior rodada (#30) Acurácia: 0.6452
```

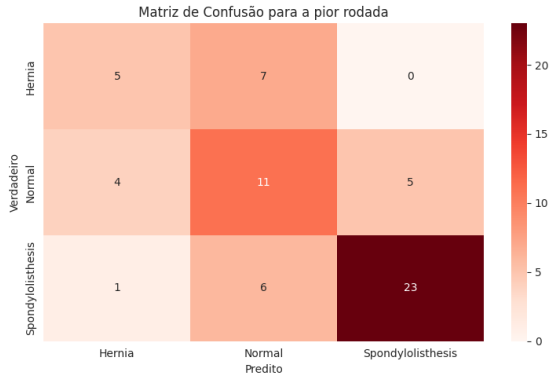





```
Classification Report for Best Run (#92):
                   precision    recall  f1-score   support

           Hernia       0.70      0.58      0.64        12
           Normal       0.78      0.90      0.84        20
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.87        62
        macro avg       0.83      0.82      0.82        62
     weighted avg       0.87      0.87      0.87        62
```

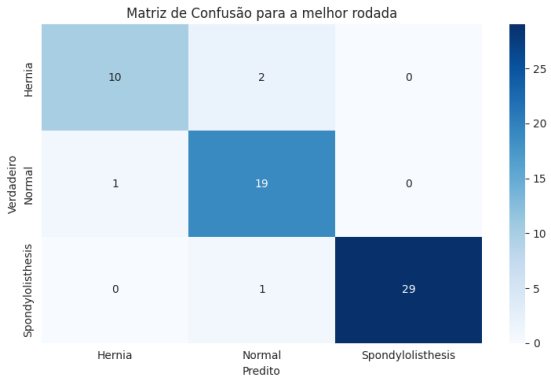
```
--- KNN (m=0.67) ---
Melhor rodada (#47) Acurácia: 0.9032
Pior rodada (#30) Acurácia: 0.6290
```

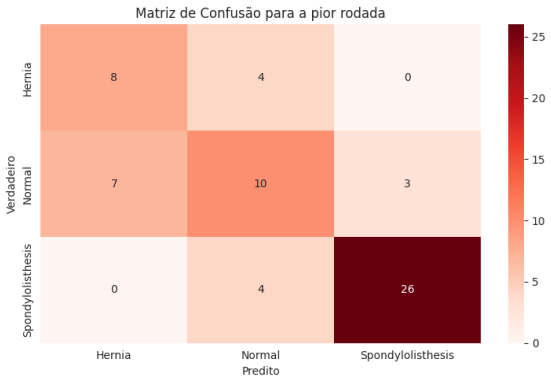




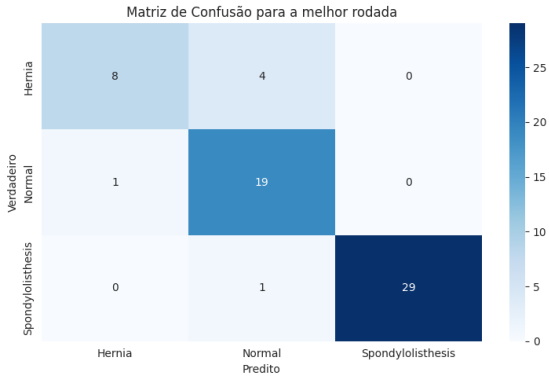

```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.79      0.92      0.85        12
           Normal       0.86      0.90      0.88        20
```

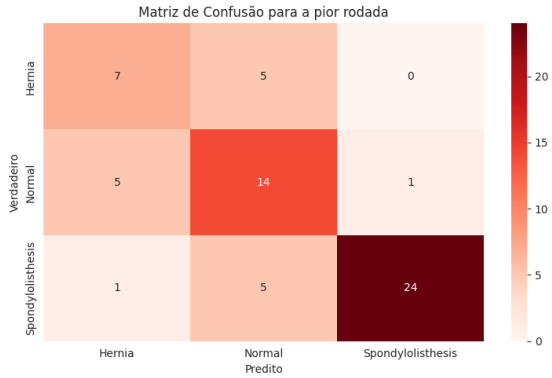
```
Spondylolisthesis       1.00      0.90      0.95        30

         accuracy                           0.90        62
        macro avg       0.88      0.91      0.89        62
     weighted avg       0.91      0.90      0.91        62

--- KNN (m=1.00) ---
Melhor rodada (#47) Acurácia: 0.9355
Pior rodada (#96) Acurácia: 0.7097
```



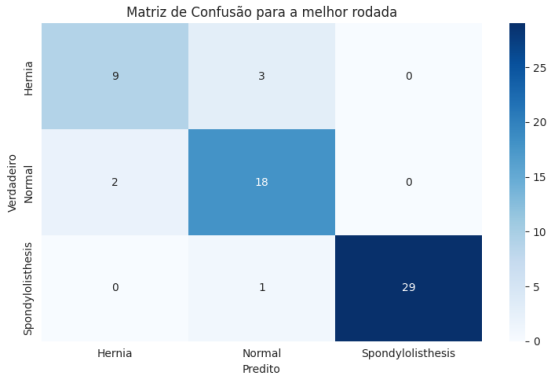


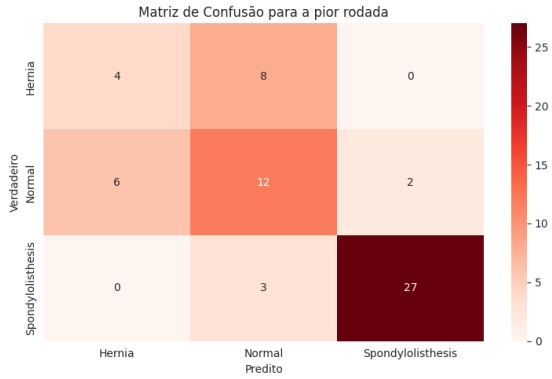

```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.91      0.83      0.87        12
           Normal       0.86      0.95      0.90        20
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.94        62
        macro avg       0.92      0.92      0.92        62
     weighted avg       0.94      0.94      0.94        62

--- KNN (m=1.50) ---
Melhor rodada (#55) Acurácia: 0.9032
Pior rodada (#29) Acurácia: 0.7258
```





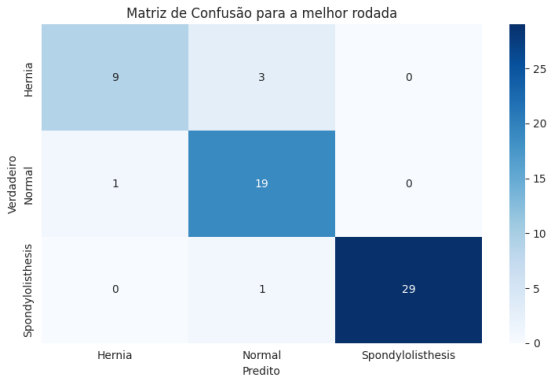

```
Classification Report for Best Run (#55):
                   precision    recall  f1-score   support

           Hernia       0.89      0.67      0.76        12
           Normal       0.79      0.95      0.86        20
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.90        62
        macro avg       0.89      0.86      0.87        62
     weighted avg       0.91      0.90      0.90        62

--- KNN (m=2.00) ---
Melhor rodada (#4) Acurácia: 0.9032
Pior rodada (#32) Acurácia: 0.6935
```

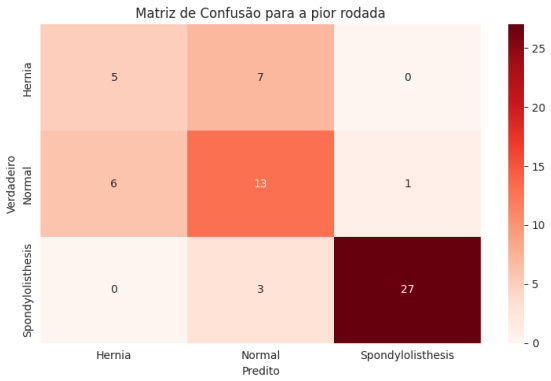




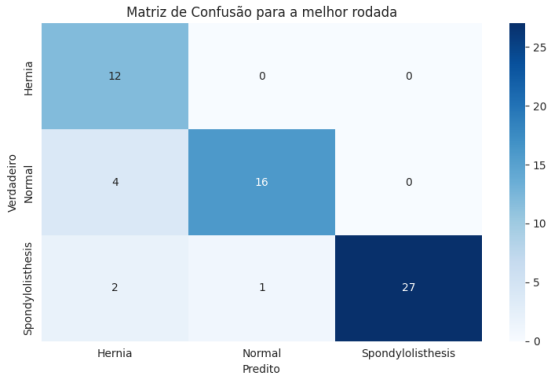

```
Classification Report for Best Run (#4):
                   precision    recall  f1-score   support

           Hernia       0.82      0.75      0.78        12
           Normal       0.82      0.90      0.86        20
```

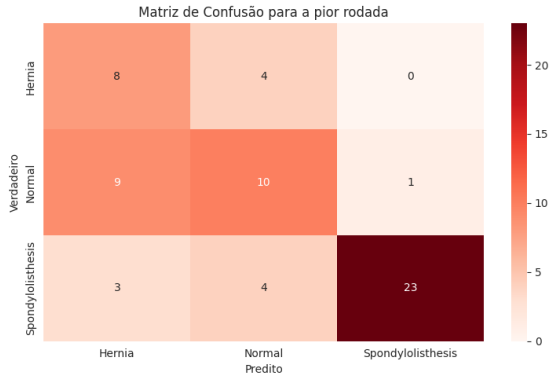
```
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.90        62
        macro avg       0.88      0.87      0.87        62
     weighted avg       0.91      0.90      0.90        62

--- KNN (m=2.50) ---
Melhor rodada (#55) Acurácia: 0.9194
Pior rodada (#32) Acurácia: 0.7258
```



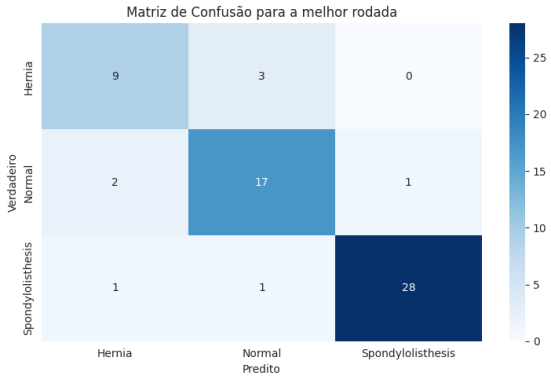


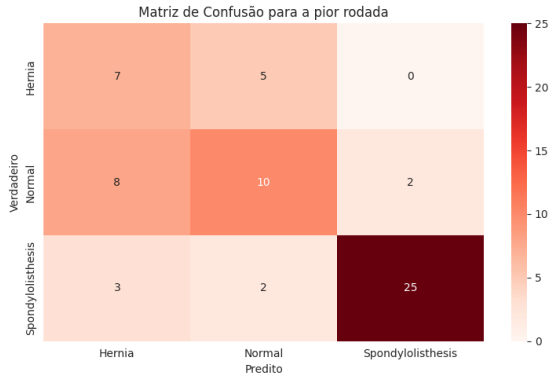

```
Classification Report for Best Run (#55):
                   precision    recall  f1-score   support

           Hernia       0.90      0.75      0.82        12
           Normal       0.83      0.95      0.88        20
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.92        62
        macro avg       0.91      0.89      0.89        62
     weighted avg       0.92      0.92      0.92        62

--- MDC ---
Melhor rodada (#23) Acurácia: 0.8871
Pior rodada (#13) Acurácia: 0.6613
```





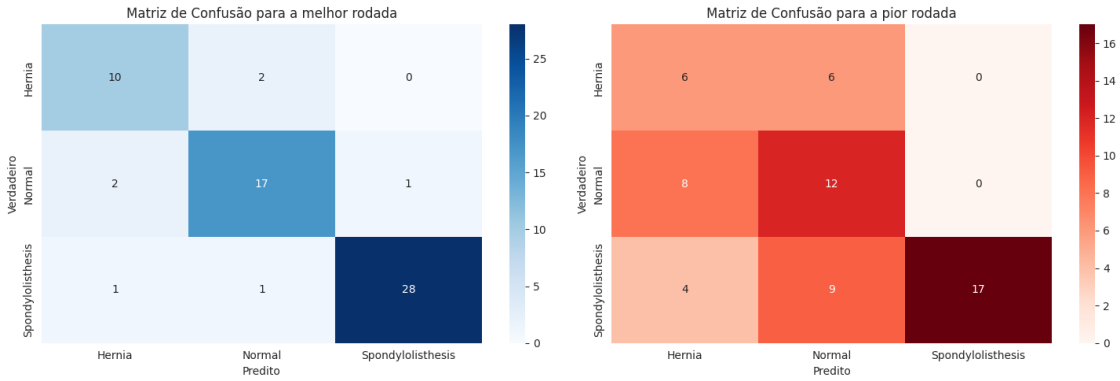

```
Classification Report for Best Run (#23):
                   precision    recall  f1-score   support

           Hernia       0.67      1.00      0.80        12
           Normal       0.94      0.80      0.86        20
Spondylolisthesis       1.00      0.90      0.95        30

         accuracy                           0.89        62
        macro avg       0.87      0.90      0.87        62
     weighted avg       0.92      0.89      0.89        62

--- MDC Robusto ---
Melhor rodada (#28) Acurácia: 0.8710
Pior rodada (#13) Acurácia: 0.6774
```






```
Classification Report for Best Run (#28):
                   precision    recall  f1-score   support

           Hernia       0.75      0.75      0.75        12
           Normal       0.81      0.85      0.83        20
```

```
Spondylolisthesis       0.97      0.93      0.95        30

        accuracy                            0.87        62
       macro avg        0.84      0.84      0.84        62
    weighted avg        0.87      0.87      0.87        62

--- MAXCO ---
Melhor rodada (#79) Acurácia: 0.8871
Pior rodada (#67) Acurácia: 0.5645
```




```
Classification Report for Best Run (#79):
                   precision    recall  f1-score   support

           Hernia       0.77      0.83      0.80        12
           Normal       0.85      0.85      0.85        20
Spondylolisthesis       0.97      0.93      0.95        30

        accuracy                            0.89        62
       macro avg        0.86      0.87      0.87        62
    weighted avg        0.89      0.89      0.89        62
```

1.0.5 Comparativo do desempenho dos algoritmos por proporção de dados de treinamento/teste

[8]:
```
resultados_df.groupby(['Classifier', 'Train Ratio']).agg({
    'Acuracia Media': 'first',
    'Tempo de Treino Medio (ms)': 'first',
    'Tempo de Treino Std (ms)': 'first',
    'Tempo de Teste Medio (ms)': 'first',
    'Tempo de Teste Std (ms)': 'first'
}).rename(columns={
    'Acuracia Media': 'Acurácia',
    'Tempo de Treino Medio (ms)': 'Treino (ms)',
```

```
    'Tempo de Treino Std (ms)': 'Treino Std (ms)',
    'Tempo de Teste Medio (ms)': 'Teste (ms)',
    'Tempo de Teste Std (ms)': 'Teste Std (ms)'
})
```

[8]:

Classifier	Train Ratio	Acurácia	Treino (ms)	Treino Std (ms)	Teste (ms)	\
KNN (m=0.50)	20	0.765484	0.003431	0.000747	0.030506	
	30	0.765699	0.003502	0.000435	0.028840	
	50	0.766774	0.003555	0.000506	0.026328	
	70	0.760276	0.003626	0.000542	0.021929	
	80	0.756815	0.003850	0.000868	0.020143	
KNN (m=0.67)	20	0.784516	0.002534	0.000648	0.049910	
	30	0.780860	0.002508	0.000240	0.045318	
	50	0.779613	0.002525	0.000296	0.038298	
	70	0.771935	0.002632	0.000371	0.029279	
	80	0.765847	0.002799	0.000591	0.025439	
KNN (m=1.00)	20	0.815806	0.002530	0.000783	0.029088	
	30	0.805699	0.002475	0.000347	0.027919	
	50	0.797677	0.002503	0.000594	0.025748	
	70	0.786498	0.002584	0.000785	0.021595	
	80	0.778710	0.002763	0.000727	0.020113	
KNN (m=1.50)	20	0.816129	0.002348	0.000244	0.049399	
	30	0.807849	0.002387	0.000777	0.045197	
	50	0.804065	0.002434	0.000329	0.038218	
	70	0.792488	0.002477	0.000290	0.029229	
	80	0.784516	0.002701	0.000761	0.025363	
KNN (m=2.00)	20	0.820968	0.002372	0.000266	0.029628	
	30	0.812151	0.002480	0.000777	0.027775	
	50	0.806581	0.002403	0.000407	0.025554	
	70	0.794931	0.002575	0.000738	0.021472	
	80	0.784516	0.002615	0.000628	0.019709	
KNN (m=2.50)	20	0.823871	0.002391	0.000269	0.049257	
	30	0.814409	0.002437	0.000718	0.045346	
	50	0.806194	0.002384	0.000380	0.038123	
	70	0.793641	0.002468	0.000388	0.029190	
	80	0.783669	0.002558	0.000533	0.025053	
MAXCO	20	0.722097	0.057039	0.004860	0.037629	
	30	0.722043	0.055487	0.003100	0.036736	
	50	0.723484	0.053742	0.006205	0.036694	
	70	0.719263	0.050995	0.003827	0.036215	
	80	0.713024	0.049694	0.004752	0.036468	
MDC	20	0.762903	0.068069	0.006299	0.005747	
	30	0.765914	0.070791	0.008751	0.005554	
	50	0.770645	0.067036	0.004947	0.005491	
	70	0.765115	0.066926	0.005259	0.005392	
	80	0.760565	0.065658	0.007431	0.005409	

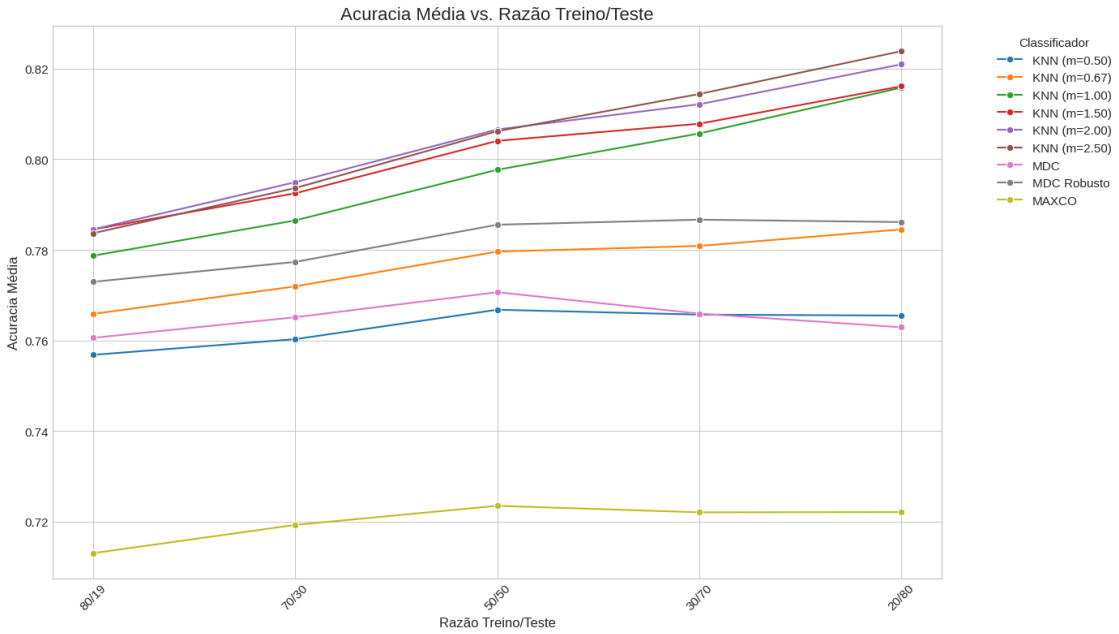
MDC Robusto	20	0.786129	0.139830	0.014982	0.005553
	30	0.786667	0.139074	0.010912	0.005531
	50	0.785548	0.132589	0.006075	0.005476
	70	0.777327	0.129135	0.007447	0.005364
	80	0.772944	0.125999	0.010985	0.005366

Classifier	Train Ratio	Teste Std (ms)
KNN (m=0.50)	20	0.001628
	30	0.000602
	50	0.000596
	70	0.000493
	80	0.001590
KNN (m=0.67)	20	0.002270
	30	0.001132
	50	0.001188
	70	0.000478
	80	0.002218
KNN (m=1.00)	20	0.001402
	30	0.001259
	50	0.000827
	70	0.000296
	80	0.001930
KNN (m=1.50)	20	0.002059
	30	0.000940
	50	0.001106
	70	0.000441
	80	0.002210
KNN (m=2.00)	20	0.002188
	30	0.001308
	50	0.000753
	70	0.000593
	80	0.001248
KNN (m=2.50)	20	0.002637
	30	0.001267
	50	0.000621
	70	0.000469
	80	0.001216
MAXCO	20	0.002178
	30	0.001013
	50	0.001032
	70	0.001371
	80	0.002619
MDC	20	0.000801
	30	0.000502
	50	0.000193
	70	0.000147

```
             80                0.000365
MDC Robusto  20                0.000334
             30                0.000443
             50                0.000139
             70                0.000097
             80                0.000243
```

1.0.6 5. OBS 3: Avaliação do Efeito da Separação Treino-Teste

[12]:
```
split_analysis_data = []
for split_ratio, results in all_split_results.items():
    for name, runs in results.items():
        mean_accuracy = np.mean([r['accuracy'] for r in runs])
        split_analysis_data.append({'Split': split_ratio, 'Classifier': name,␣
 ↪'Mean Accuracy': mean_accuracy})

split_df = pd.DataFrame(split_analysis_data)

plt.style.use('seaborn-v0_8-whitegrid')
fig, ax = plt.subplots(figsize=(14, 8))
sns.lineplot(data=split_df, x='Split', y='Mean Accuracy', hue='Classifier',␣
 ↪marker='o', ax=ax)
ax.set_title('Acuracia Média vs. Razão Treino/Teste', fontsize=16)
ax.set_xlabel('Razão Treino/Teste', fontsize=12)
ax.set_ylabel('Acuracia Média', fontsize=12)
ax.legend(title='Classificador', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



[10]:
```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd
from ucimlrepo import fetch_ucirepo

# Configurações
sns.set_style("whitegrid")
plt.rcParams.update({'font.size': 11})

# Carregamento (caso não esteja carregado)
vertebral_column = fetch_ucirepo(id=212)
df = vertebral_column.data.features.copy()
df['class'] = vertebral_column.data.targets.to_numpy().ravel()

# ---. Cálculo dos Centróides (Para argumentar sobre o MDC) ---
# Vamos calcular e exibir a distância entre os centróides para provar a␣
↪separabilidade
print("\n--- Análise dos Centróides (Hipótese MDC) ---")
centroids = df.groupby('class').mean()
display(centroids)

# Dica de Mestre: Calcule a distância Euclidiana entre os centróides.
# Se a distância entre 'Hernia' e 'Normal' for pequena comparada à variância␣
↪interna (dispersão),
# o MDC vai falhar miseravelmente nessas duas classes.
```

```python
from scipy.spatial.distance import cdist
dists = pd.DataFrame(
    cdist(centroids, centroids, metric='euclidean'),
    index=centroids.index,
    columns=centroids.index
)
print("\nMatriz de Distâncias Euclidianas entre Centróides:")
display(dists.style.background_gradient(cmap='Blues'))
```

```
--- Análise dos Centróides (Hipótese MDC) ---

                   pelvic_incidence  pelvic_tilt  lumbar_lordosis_angle  \
class
Hernia                    47.638407    17.398795              35.463524
Normal                    51.685244    12.821414              43.542605
Spondylolisthesis         71.514224    20.748038              64.110108

                   sacral_slope  pelvic_radius  degree_spondylolisthesis
class
Hernia                30.239612     116.474968                  2.480251
Normal                38.863830     123.890834                  2.186572
Spondylolisthesis     50.766186     114.518810                 51.896687

Matriz de Distâncias Euclidianas entre Centróides:

<pandas.io.formats.style.Styler at 0x7bf641382300>
```